

Player positions and the 4×4 Q matrix

I examine fourteen representative soccer states. Each is identified by its state, five numbers (x_0 , y_0 , x_1 , y_1 , b): player 0's cell (x_0 , y_0), player 1's cell (x_1 , y_1), and b (which player currently has the ball, 0 or 1). That five-number state is not part of the Q matrix — it just names *which* stage game is shown; the Q matrix itself is the 4×4 table of payoffs indexed by the two players' *actions* (U D L R each), not by coordinates.

For each state, the **left panel** shows the physical player configuration and every action either player actually takes in equilibrium, drawn as probability arrows (thick = likely, thin = unlikely, no arrow at all = zero probability). The **right panel** shows the complete 4×4 stage-game Q matrix — the payoff table $Q(s, a_0, a_1)$ at that one state, not the value function $V(s)$ — redrawn as a node-and-arrow graph. Always the full grid, never reduced. The goal is not simply to show *where* mixing happens on the board, but **what the players are actually doing when it does**.

A player's **support** is the set of actions it assigns positive probability in an equilibrium strategy — support (2,2) means the carrier mixes over 2 actions and the defender mixes over 2 actions.

Canonical support shapes

CARRIER SUPPORT	DEFENDER SUPPORT	SHAPE	STATES
2 actions	2 actions	(2,2)	68
3 actions	3 actions	(3,3)	4
2 actions	1 action	(2,1)	14
3 actions	2 actions	(3,2)	8
total no-pure-saddle states			94

Every case below can be opened directly in the [interactive explorer](#) at its own position. For the deterministic A10 board's own edge cases (every stage game there is provably pure, but ties and the collision rule still hide real structure), see [a10_cases.pdf](#); the tackle, territory, and movement-slip rule variants touched on in cases 8, 11, and 12 below each get their own dedicated case-study document too ([tackle_cases.pdf](#), [territory_cases.pdf](#), [slip_cases.pdf](#)), alongside the coin-flip contest-resolution variant ([coinflip_cases.pdf](#)).

The defender's support is never larger than the carrier's, in any of the 94 no-pure-saddle states on the canonical board.

The bigger finding underneath those four rows: 94 no-pure-saddle states are not 94 different phenomena — they are four repeated shapes, recurring all over the board.

Three kinds of fractional output, labelled below on every case

A reported fractional split is not always a number the game forces. Every case below is labelled with exactly one of these (`docs/degeneracy.md` has the full breakdown, all 94 states):

TYPE	MEANING	COUNT
Forced mixed equilibrium	neither side has an unweighted action tied with its reported support's value — the split shown is the whole indifference class, no freedom left over.	64
Degenerate equilibrium face	a side's reported support already has two or more actions, and a further, unweighted action ties them exactly — one vertex of a larger polytope, not a uniquely forced ratio.	16
Pure-tied / fractional LP output	a side's reported support is a single action (nominally “pure”), but an unweighted action ties it exactly — just as arbitrary as a printed fractional split.	14

Case 7 shows one of the 64 repeats directly (Case 2's exact shape, elsewhere on the board, mirrored). This page shows one example of each of the four canonical support shapes, plus every distinct mechanism studied in the project that can force a mix: a stochastic move order, this project's own tackle rule, a dense reward with zero transition noise, and movement slip on a goal shape that is otherwise always pure. Cases 13 and 14 complete a stronger claim: docs/templates.md reduces the 94 no-pure-saddle states to **8 canonical geometric templates**, and with those two, every one of the 8 now has its own full case here too (template $\rightarrow$ case: 1 $\rightarrow$ 4, 2 $\rightarrow$ 2, 3 $\rightarrow$ 13, 4 $\rightarrow$ 9, 5 $\rightarrow$ 10, 6 $\rightarrow$ 5, 7 $\rightarrow$ 3, 8 $\rightarrow$ 14). Every one of the fourteen gets its own full-size board-and-matrix figure below — no combined overview image, so nothing here is ever cropped by a page break.

Reading the board and the Q matrix

The board (left). Each player is a filled circle; the ball sits on whoever carries it. An arrow points from a player toward a cell it might move to, and its **thickness and opacity are the equilibrium probability** of that action — a bold arrow is very likely, a thin one is unlikely, and an action with (numerically) zero probability draws no arrow at all. A pure policy is one bold arrow and nothing else; a mixed policy fans out into two or three arrows of different weights.

The Q matrix (right). Each node is one cell of the complete 4 $\times$ 4 stage matrix (row = player 0's action, col = player 1's action, **fixed for every state**, never reoriented by who currently has the ball — the payoff shown is always player 0's). A blue arrow points from a cell to the cell in the same column with the higher payoff — player 0's own reason to switch rows, since player 0 maximises. A green arrow points from a cell to the cell in the same row with the *lower* payoff — player 1's reason to switch columns, since player 1 minimises. Which player is carrying the ball at a given state is stated separately under each case; it does not change which row or column is which. Every case shows all 16 cells, so the shapes are directly comparable.

This is the mathematical justification *for* what the board already shows: it is why the players' support is exactly what it is.

A pure state has exactly one node with no outgoing arrow — that cell is the saddle, everything else eventually points to it. A mixed state has no such node: no single joint action is simultaneously stable for both players. At the equilibrium, every action either player assigns positive probability to gives that player the *same* expected payoff against the opponent's actual mix (worked out explicitly under “why indifferent” for each case below) — that is the substance of a mixed equilibrium, not merely that the arrows visually cycle.

A third piece, below the board and the Q matrix: the successor-state coordinates. Every case also gets a 4×4 table, same orientation as the Q matrix, whose *cells* are the actual (x_0 , y_0 , x_1 , y_1 , b) state reached by each joint action — what `game.transitions()` returns, not what the payoff is. Where the transition is stochastic (random move order, or the tackle rule), a cell lists both possible successors and their probabilities. This is the direct answer to “where do the numbers in the Q matrix actually come from”: each payoff is the expected value over exactly these successor states.

A wall clamp is a hold, not a move. A few of the states below have a player already against an edge of the board — attempting to move further that way (U at the top row, D at the bottom, L at the left column, R at the right) does not move the player at all; it silently returns the same cell. Where the equilibrium puts real weight on one of these no-ops, the board figure draws it as a dashed **hold** ring around the player (the same treatment as an explicit STAND action), not a directional arrow — an arrow pointing into a wall would show movement that never happens. Cases 4, 5, 6, 8, 10 and 11 below each have at least one player pinned this way; each is called out explicitly where it matters.

Every number on this page is exact, not an iterative approximation.

`NashQIteration.run()` finds the equilibrium policy at every state by sweeping until the largest update drops below a tolerance — by construction, an approximation to the true fixed point, however tight. Once that policy (which states are a pure saddle, and the LP mixed strategy where

they are not) is known, it induces a fixed Markov chain over states, and the value function is the *unique* solution of the linear system $V = R + \gamma P @ V$. Solving that system directly, in one sparse linear solve instead of sweeping toward a tolerance, is exactly what `NashQIteration.run_exact()` does, and every Q matrix below is built from that exact V. The two never disagreed by more than $2.3e-10$ on the canonical board ($4.2e-10$ on the slip board, $8.2e-10$ on the territory-reward board, $3.8e-11$ on the tackle board) — a direct, numeric certificate that the iterative solver had actually converged, not just a claim about the tolerance it was run with.

Five representative cases

For a focused read: the typical two-action mix, the cleanest indifference example, a genuine three-action mix, the sharpest LP-degeneracy example, and a case where a completely different mechanism (the reward, not the transition) forces the mix. All fourteen cases, these five included, follow in full below in numbered order.

Case 2 (0, 1, 1, 1, 0)

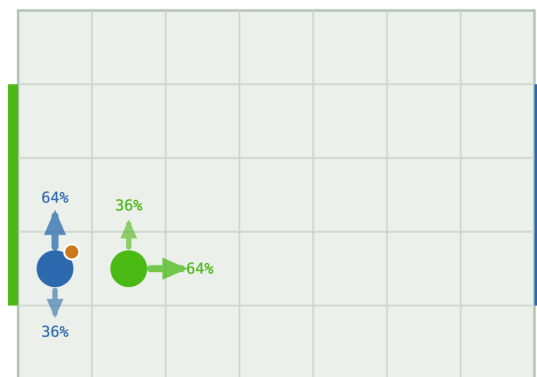
The primary example — the typical mix. Gap 0.0136. The carrier mixes U 63.5% / D 36.5%; the defender answers U 36.5% / R 63.5%. This is not one strange state among many — it is the **dominant pattern in the entire mixed-state set: 90 of the 94 no-pure-saddle states on the canonical board cross a vertical pair exactly like this one**, the carrier deciding which goal row to attack and the defender guessing which one. Everything that follows is either this same shape recurring (Case 7), or a genuinely different shape worth contrasting against it (Cases 3, 4, 5, 9, 10).

Type: forced mixed equilibrium — the reported 63.5%/36.5% split is the whole indifference class, not one point on a larger face.

Support: carrier {U, D} (63.5% / 36.5%) · defender {U, R} (36.5% / 63.5%)

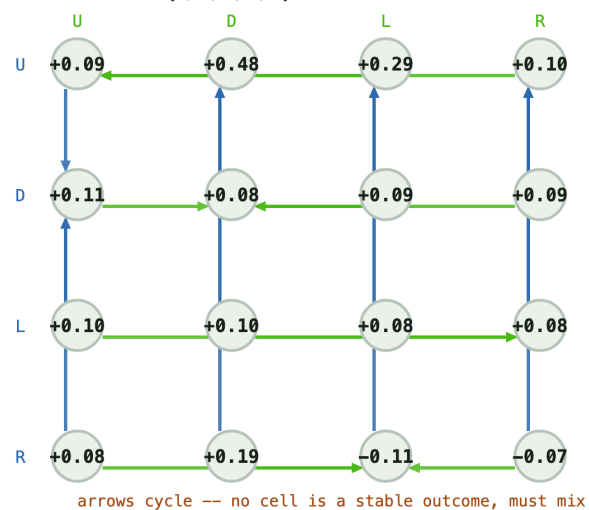
Why indifferent: Carrier $E[U] = E[D] = +0.094$, strictly above $E[L] = +0.091$ and $E[R] = -0.015$. Defender $E[U] = E[R] = +0.094$, strictly below $E[L] = +0.212$ and $E[D] = +0.332$.

The primary example: the typical mix



mixed equilibrium · $V = +0.094$

Q matrix -- (0, 1, 1, 1, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 1, 1, 1, 0)
Carrier	player 0 at (0, 1)
Defender	player 1 at (1, 1)

Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.085599	0.478297	0.285670	0.099198
D	0.109260	0.076330	0.085331	0.085599
L	0.103381	0.103381	0.084811	0.083738
R	0.082480	0.187381	-0.108448	-0.070696

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 63.5% / D 36.5%	U 36.5% / R 63.5%
3 decimal(s)	mixed	U 63.9% / D 36.1%	U 36.1% / R 63.9%
2 decimal(s)	mixed	U 66.7% / D 33.3%	U 33.3% / R 66.7%
1 decimal(s)	pure	L 100.0%	U 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(0,2,1,2,0)	(0,2,1,0,0)	50%: (0,2,0,1,0) 50%: (0,2,1,1,0)	(0,2,2,1,0)
D	(0,0,1,2,0)	(0,0,1,0,0)	50%: (0,0,0,1,0) 50%: (0,0,1,1,0)	(0,0,2,1,0)
L	(0,1,1,2,0)	(0,1,1,0,0)	(0,1,1,1,0)	(0,1,2,1,0)
R	50%: (0,1,1,2,1) 50%: (1,1,1,2,0)	50%: (0,1,1,0,1) 50%: (1,1,1,0,0)	50%: (0,1,1,1,0) 50%: (0,1,1,1,1)	50%: (0,1,2,1,1) 50%: (1,1,2,1,0)

Case 3 (1, 1, 1, 0, 1)

A clean L/R indifference example. Placed directly after the primary example on purpose: this is the cleanest demonstration of the intuition Case 2 only implies —

"when we move right and when we move left, there's an equal chance of me winning. That's why I am indifferent between the two."

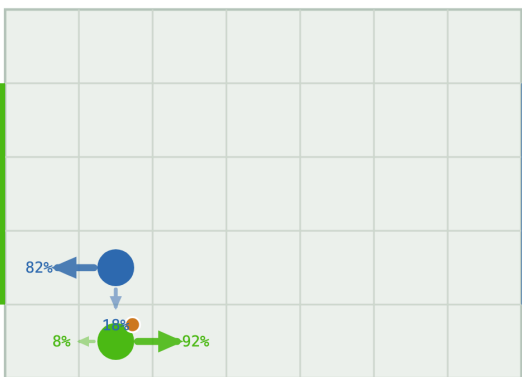
Gap 0.0445. The carrier's entire live option set is exactly $\{L, R\}$ — no vertical option survives at all, which is what makes this state (rather than Case 4, which crosses U/L) the cleanest match to that description: two actions genuinely equally good against the defender's own mix. **This is the rarer shape overall: only 4 of 94 no-pure-saddle states cross a purely horizontal pair instead of a vertical one.**

Type: forced mixed equilibrium — both sides' splits are the whole indifference class.

Support: carrier $\{L, R\}$ (7.7% / 92.3%) · defender $\{D, L\}$ (18% / 82%)

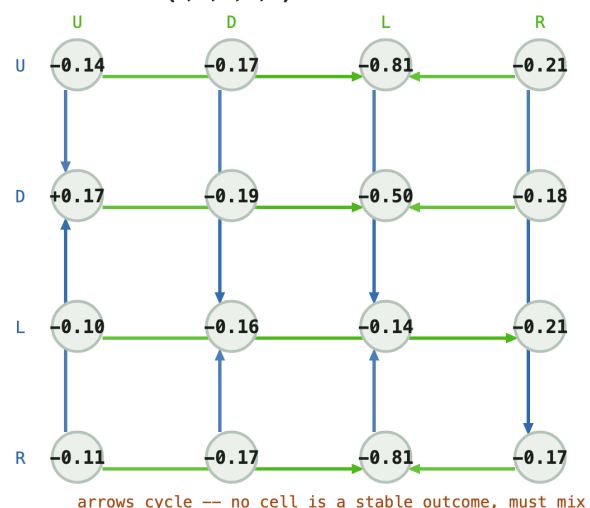
Why indifferent: Carrier $E[L]=E[R]=+0.206$, strictly above $E[D]=+0.162$ and $E[U]=+0.050$. Defender $E[D]=E[L]=+0.206$, strictly below $E[R]=+0.216$ and $E[U]=+0.257$.

A clean L/R indifference example



mixed equilibrium · $V = -0.206$

Q matrix -- (1, 1, 1, 0, 1)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(1, 1, 1, 0, 1)
Carrier	player 1 at (1, 0)
Defender	player 0 at (1, 1)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value — fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	-0.139279	-0.170790	-0.810000	-0.211001
D	0.173204	-0.185032	-0.499883	-0.180940
L	-0.099163	-0.156678	-0.141010	-0.211001
R	-0.109755	-0.170790	-0.810000	-0.166529

Where two of these cells actually come from — every entry is $Q(s, a_0, a_1) = \text{sum_outcomes } P(\text{outcome}) * (r_0 + \text{gamma} * V(\text{next}))$, real game.transitions() outcomes and the solver's own V, not the matrix solve itself. Two worked examples, $\text{gamma} = 0.9$:

$Q(U, U) = -0.139279$ — off both players' actual support, a bad joint action for player 0:

OUTCOME	P	NEXT	R0	V(NEXT)	P*(R0 + Γ *V(NEXT))
move first	0.5	(1, 2, 1, 1, 1)	0	-0.900000	-0.405000
move second	0.5	(1, 2, 1, 0, 0)	0	+0.590490	+0.265721
sum					-0.139279

$Q(D, U) = +0.173204$ — inside the defender's real 18% support, and notable because one of its two successors is the state itself:

OUTCOME	P	NEXT	R0	V(NEXT)	P*(R0 + Γ *V(NEXT))
move first	0.5	(1, 1, 1, 0, 0)	0	+0.590490	+0.265721
move second	0.5	(1, 1, 1, 0, 1) (= s itself)	0	-0.205592	-0.092516
sum					+0.173204

Neither cell has a per-step reward ($r_0 = 0$): the "win" objective only pays ± 1 on the step a goal is actually scored, so every non-terminal cell's value comes entirely from $\gamma * V(\text{next})$, discounted by whichever successor's own certified value it lands on — including, for $Q(D, U)$, this very state's own $V = -0.205592$, since D/U can return play to $(1, 1, 1, 0, 1)$ unchanged. (Cross-checked against `game.transitions()` and `result.values` directly in `tests/test_positions_verification.py`, not hand-derived.)

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0 (DEFENDER)	PLAYER 1 (CARRIER)
full (unrounded)	mixed, gap 0.0445	D 18.0% / L 82.0%	L 7.7% / R 92.3%
3 decimals	mixed	D 18.0% / L 82.0%	L 7.7% / R 92.3%
2 decimals	mixed	D 17.9% / L 82.1%	L 7.7% / R 92.3%
1 decimal	pure	L 100%	R 100%

Two decimals reproduces the exact mix to within a percentage point. One decimal manufactures a pure saddle — value exactly -0.2000 , defender pure L, carrier pure R — that is not actually there: it is an artifact of the rounding, not a feature of the game. There is no single mathematically privileged rounding precision; this table exists so that question is checkable rather than assumed one way or the other (`soccer_nash.numerics.rounding_diagnostic`).

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x_0, y_0, x_1, y_1, b) . Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	50%: (1,2,1,1,1) 50%: (1,2,1,0,0)	(1,2,1,0,1)	(1,2,0,0,1)	(1,2,2,0,1)
D	50%: (1,1,1,0,0) 50%: (1,1,1,0,1)	(1,1,1,0,1)	50%: (1,1,0,0,1) 50%: (1,0,0,0,1)	50%: (1,1,2,0,1) 50%: (1,0,2,0,1)
L	50%: (0,1,1,1,1) 50%: (0,1,1,0,0)	(0,1,1,0,1)	(0,1,0,0,1)	(0,1,2,0,1)
R	50%: (2,1,1,1,1) 50%: (2,1,1,0,0)	(2,1,1,0,1)	(2,1,0,0,1)	(2,1,2,0,1)

Why not defend by moving right instead? The natural reading of this state — player 1 has the ball right next to player 0 — is "close the distance," which on this board means R, not L. That reading misses one board detail: `goal_rows = (1, 2, 3)` on a 5-tall board, so row **y=0**, where the carrier currently stands, is not a scoring row at all — reaching $x=-1$ from $y=0$ does not score. The carrier is not a live threat yet; it first has to climb into a goal row before a leftward run means anything, and the cell it needs is $(0, 1)$, one step from the defender's own current cell $(1, 1)$ by playing L. The defender's job here is to deny that *future* cell, not to react to the carrier's *current* one.

The cost of getting this backwards is not subtle. Taking the worst case over the carrier's reply to each pure defender action, straight from the Q matrix above:

DEFENDER PLAYS	WORST CASE FOR DEFENDER	CARRIER'S BEST REPLY	SUCCESSOR STATE
U	-0.810	L	$(1,2,0,0,1)$ -- defender at $(1,2)$, corner wide open
R	-0.810	L	$(2,1,0,0,1)$ -- defender moves away, same open corner
L	-0.211	R	$(0,1,0,0,1)$ -- defender now sits on the corner itself

U and R are not merely worse than L, they are close to the worst possible outcome on this board — a goal is worth ± 1 , and -0.81 under $\gamma=0.9$ is what "the opponent is about to score" looks like in this game's own value scale. Playing L is not a retreat: the defender ends up standing on $(0, 1)$, exactly the cell the carrier would otherwise have walked into uncontested.

Once that cell is covered, the carrier's own best reply shifts away from L toward R (92.3% of the mix) — not because R is a good attacking move on its own, but because L only pays off if $(0, 1)$ is left open, and the defender's 82% weight on L makes sure it usually is not.

This entire case depends on the goal convention. Everything above relies on row $y=0$ (where the carrier starts) not being a goal row under this project's `goal_rows = (1, 2, 3)`. Under a full-boundary goal (`goal_rows = (0, 1, 2, 3, 4)`, every row goal-eligible) this exact state is **pure, not mixed**: the carrier's L wins in two moves regardless of the defender's reply, all four of the defender's actions tie at the same value, and there is no guessing game left to play at all — see [generalize.md](#) for the numbers. `goal_rows = (1, 2, 3)` is Littman's own convention and this repo's default, confirmed as the better choice for this project, but Case 3 specifically — not just the aggregate mixed-state count — would need to be re-derived under a different one.

Case 5

(0, 0, 2, 0, 0)

A **genuine 3-action mix**. Six cells from goal, as far as this board allows, with the defender close enough to threaten three different rows at once: no single row is safely better than the other two, so three actions are simultaneously undominated on both sides. **This is one of only 4 states on the whole canonical board with support shape (3,3)** — mixing is not always a 50/50 split between two actions. (Entropy 1.109 bits, the richest mix on this page, if a single summary number is wanted — but the three-way split itself is the point.)

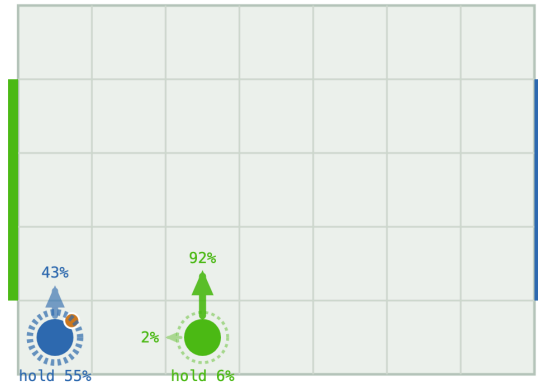
Both players are pinned against an edge here, so read the raw action letters with that in mind: the carrier sits at (0, 0), so L is a wall no-op — it clamps straight back to (0, 0), physically identical to D — which is why the board figure draws that 54.7% as a dashed **hold** ring, not a leftward arrow. The three genuinely distinct carrier choices are *attack up* (U, 43.3%), *hold ground* (D/L combined, 54.7%), and *sneak right* (R, 1.9%) — still a real three-way mix, just not the three-arrows picture the raw letters would suggest. The defender at (2, 0) has the same issue in miniature: D is its own wall no-op (5.9%, folded into its own hold ring), alongside real moves U (91.9%) and L (2.3%).

Type: degenerate equilibrium face (carrier) — a fourth, unweighted carrier action ties the reported three-way split exactly (see “why indifferent”), so the printed split is one point on a larger face, not a forced ratio. Here that tie is not a coincidence of the LP: D and L are the *same physical action* from (0, 0), so $E[D]=E[L]$ is a mechanical certainty, not a numerical accident. The defender's own split is a genuine forced mix.

Support: carrier {U, L, R} (43.3% / 54.7% / 1.9%, L a wall hold) · defender {U, D, L} (91.9% / 5.9% / 2.3%, D a wall hold)

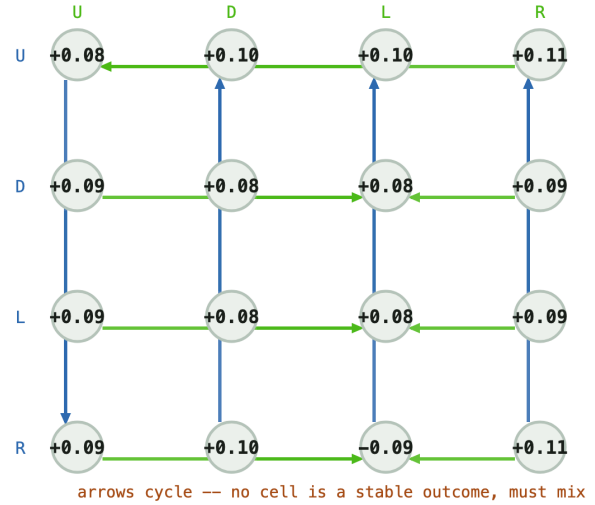
Why indifferent: Carrier $E[U]=E[L]=E[R]=+0.0848$ (the three actions weighted) — and $E[D]=+0.0848$ too: **all four actions tie exactly**, and $E[D]=E[L]$ here is guaranteed before any LP runs, since D and L are the same wall-clamped no-op from (0, 0). The printed 43.3%/0%/54.7%/1.9% split is one point on a whole face of equally-good carrier mixes, not a uniquely forced ratio. Defender $E[U]=E[D]=E[L]=+0.0848$, below $E[R]=+0.097$.

A genuine 3-action mix, not a 2-cycle



mixed equilibrium · $V = +0.085$

Q matrix -- (0, 0, 2, 0, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 0, 2, 0, 0)
Carrier	player 0 at (0, 0)
Defender	player 1 at (2, 0)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.083738	0.095109	0.103381	0.110220
D	0.085599	0.076363	0.076330	0.085599
L	0.085599	0.076363	0.076330	0.085599
R	0.088459	0.095109	-0.088213	0.110220

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 43.3% / L 54.7% / R 1.9%	U 91.9% / D 5.9% / L 2.3%
3 decimal(s)	mixed	U 46.0% / L 52.0% / R 2.0%	U 91.2% / D 6.9% / L 1.9%
2 decimal(s)	mixed	U 33.3% / L 66.7%	U 66.7% / D 29.8% / L 3.5%
1 decimal(s)	pure	L 100.0%	L 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x_0, y_0, x_1, y_1, b) . Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	$(0, 1, 2, 1, 0)$	$(0, 1, 2, 0, 0)$	$(0, 1, 1, 0, 0)$	$(0, 1, 3, 0, 0)$
D	$(0, 0, 2, 1, 0)$	$(0, 0, 2, 0, 0)$	$(0, 0, 1, 0, 0)$	$(0, 0, 3, 0, 0)$
L	$(0, 0, 2, 1, 0)$	$(0, 0, 2, 0, 0)$	$(0, 0, 1, 0, 0)$	$(0, 0, 3, 0, 0)$
R	$(1, 0, 2, 1, 0)$	$(1, 0, 2, 0, 0)$	50%: $(1, 0, 2, 0, 0)$ 50%: $(0, 0, 1, 0, 1)$	$(1, 0, 3, 0, 0)$

Case 9 (0, 2, 1, 2, 0)

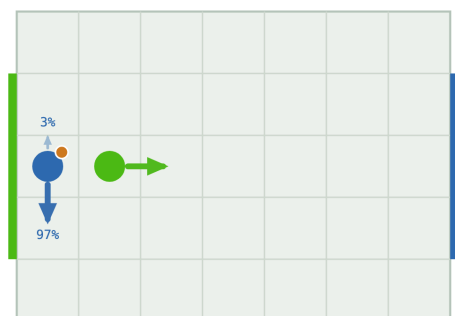
The asymmetric mix — the sharpest LP-degeneracy example. Support (2, 1)
 — gap 0.0095. The odd one out: the carrier's equilibrium still mixes two actions (U 2.6% / D 97.4%), but the **defender's** equilibrium is a single fixed move (R, 100%). Every other case on this page is a symmetric duel — both players genuinely guessing. Here only one side is.

Type: pure-tied / fractional LP output (defender) — the defender's official policy is printed pure R (100%), but D ties it exactly: the choice of R over D is exactly as arbitrary as a printed fractional split. The sharpest example on this page that a fractional LP output is not automatically a strategically required mixed strategy.

Support: carrier {U, D} (2.6% / 97.4%) · defender {R} (100%)

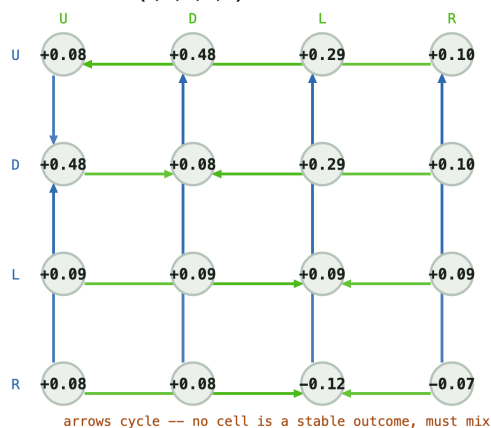
Why indifferent: Carrier $E[U] = E[D] = +0.095$, above $E[L] = +0.093$ and $E[R] = -0.072$ (worked out fully below). The defender's official policy is pure R (100%), but $E[R] = +0.095$ and $E[D] = +0.095$ tie **too**: the defender's own choice of R over D is exactly as arbitrary as the carrier's U/D split, it just isn't printed as a percentage because the solver put all the weight on one side.

The asymmetric mix -- a tie, not a forced mix, dressed the same



mixed equilibrium · $V = +0.095$

Q matrix -- (0, 2, 1, 2, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 2, 1, 2, 0)
Carrier	player 0 at (0, 2)
Defender	player 1 at (1, 2)
Rows	Player 0
Columns	Player 1

Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball
---------------	--

	U	D	L	R
U	0.084811	0.478297	0.290839	0.095109
D	0.478297	0.084811	0.290839	0.095109
L	0.093043	0.093043	0.085599	0.093043
R	0.082303	0.082303	-0.122276	-0.071591

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 2.6% / D 97.4%	R 100.0%
3 decimal(s)	mixed	U 2.5% / D 97.5%	R 100.0%
2 decimal(s)	mixed	U 5.0% / D 95.0%	R 100.0%
1 decimal(s)	pure	D 100.0%	R 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(0,3,1,3,0)	(0,3,1,1,0)	50%: (0,3,0,2,0) 50%: (0,3,1,2,0)	(0,3,2,2,0)
D	(0,1,1,3,0)	(0,1,1,1,0)	50%: (0,1,0,2,0) 50%: (0,1,1,2,0)	(0,1,2,2,0)
L	(0,2,1,3,0)	(0,2,1,1,0)	(0,2,1,2,0)	(0,2,2,2,0)
R	50%: (0,2,1,3,1) 50%: (1,2,1,3,0)	50%: (0,2,1,1,1) 50%: (1,2,1,1,0)	50%: (0,2,1,2,0) 50%: (0,2,1,2,1)	50%: (0,2,2,2,1) 50%: (1,2,2,2,0)

Why the carrier still needs two actions against a defender who never varies: against the defender's pure R, column R reads U 0.095, D 0.095, L 0.093, R -0.072 — **U and D are exactly tied**, both strictly better than L or R. Nothing in the payoffs favours one over the other, so any split of U/D is an equally valid best reply; the solver's LP reports one particular split (2.6% / 97.4%), not a probability forced by the game the way Case 3's 7.7% / 92.3% is. This is the sharpest example on this page of a general fact worth stating

plainly: a fractional LP output is not automatically a strategically required mixed strategy. A tie between two rows can look identical, in the printed policy, to a genuine forced mix, but it is a different phenomenon — no best-reply cycle is involved, and either pure U or pure D alone would also be a valid equilibrium reply to the defender's fixed R.

Case 11 (4, 4, 5, 4, 0), deterministic

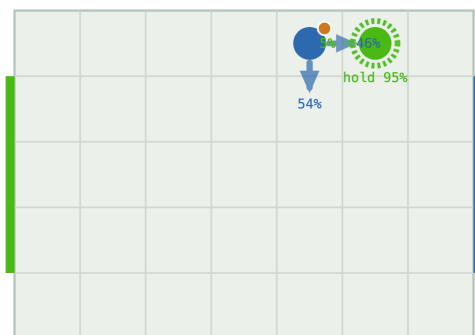
Mixing forced by reward alone, not transition. An advanced case, illustrating a fundamentally different mechanism from the four above. Every case so far gets its coin flip from a stochastic transition — the random move order. This one has **no transition stochasticity at all**: the mix comes entirely from scoring="territory", a dense per-step reward for the ball in the opponent's final third, layered on top of the ordinary win/loss score. Gap 0.0653 — close to a fair coin (entropy 0.996 bits), forced purely by coupling the reward to both players' actions. **Mixing does not require stochastic transitions; it can come from the payoff structure alone.** The defender sits at (5, 4), the top edge, so its reported majority action U (95.1%) is a wall no-op: it holds position 95.1% of the time and slides left (L) the other 4.9%.

Type: forced mixed equilibrium — reward-coupling forces a genuine indifference on both sides, the same as a transition-coupled matching-pennies core.

Support: carrier {D, R} (53.8% / 46.2%, both real moves) · defender {hold, L} (95.1% / 4.9%, printed as U/L)

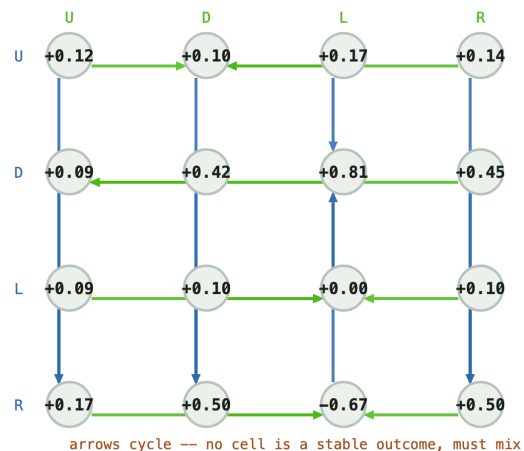
Why indifferent: Carrier $E[D] = E[R] = +0.129$, above $E[U] = +0.119$ and $E[L] = +0.090$. Defender $E[U] = E[L] = +0.129$, below $E[D] = +0.455$ and $E[R] = +0.473$ — U here is the defender holding at the top edge, tied with a real move left.

Mixing forced by reward alone -- zero transition randomness



mixed equilibrium · $V = +0.129$

Q matrix -- (4, 4, 5, 4, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(4, 4, 5, 4, 0)
Carrier	player 0 at (4, 4)
Defender	player 1 at (5, 4)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.116139	0.104525	0.169846	0.135972
D	0.094073	0.415483	0.814500	0.450000
L	0.094073	0.104525	0.000000	0.104525
R	0.169846	0.500000	-0.670721	0.500000

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	D 53.8% / R 46.2%	U 95.1% / L 4.9%
3 decimal(s)	mixed	D 53.8% / R 46.2%	U 95.1% / L 4.9%
2 decimal(s)	mixed	D 53.8% / R 46.2%	U 94.9% / L 5.1%
1 decimal(s)	mixed	D 56.2% / R 43.8%	U 93.8% / L 6.2%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(4,4,5,4,0)	(4,4,5,3,0)	(4,4,5,4,1)	(4,4,6,4,0)
D	(4,3,5,4,0)	(4,3,5,3,0)	(4,3,4,4,0)	(4,3,6,4,0)
L	(3,4,5,4,0)	(3,4,5,3,0)	(3,4,4,4,0)	(3,4,6,4,0)
R	(4,4,5,4,1)	(5,4,5,3,0)	(5,4,4,4,1)	(5,4,6,4,0)

From worked cases to exploitability: pure vs. mixed, at these exact states

Everything above stays inside one stage game — what the carrier and defender do *at* a state. `tournament_deepdive.md` shows the whole-game consequence — greedy collapses against a challenger, minimax doesn't — but at the level of the *tournament*, not any one of these states. This table connects the two directly: at each state, force the carrier to its single most likely ("greedy") action and let the defender best-respond *within that one stage game*, then compare to the actual Nash mixed value. Three rows (Cases 4, 6, 10) are detailed further down in the complete set below; the other four are the lead cases above.

CASE	GREEDY PURE VALUE VS. BR	NASH MIXED VALUE VS. BR	VALUE OF MIXING
Case 2 — the primary example	+0.0856	+0.0942	+0.0086
Case 3 — L/R indifference	+0.1665	+0.2056	+0.0391
Case 4 — the corner duel	+0.0763	+0.0776	+0.0012
Case 5 — 3-action mix	+0.0763	+0.0848	+0.0085
Case 6 — near-pure hedge	+0.1665	+0.1676	+0.0011
Case 9 — the asymmetric mix	+0.0848	+0.0951	+0.0103
Case 10 — the three-lane mix	+0.2471	+0.2816	+0.0345

The value of mixing is never negative — exactly the theoretical guarantee (an equilibrium mix is a best response to the opponent's best response; nothing pure can beat that). It is also not uniform: Case 3's clean L/R indifference is worth +0.039, fifteen times more than Case 6's near-pure hedge (+0.0011, unsurprising — a 97.5/2.5 hedge is *almost* what the greedy action already does) or Case 4's corner duel (+0.0012, the corner leaves little room to exploit either way). Case 3 and Case 10 -- the deepest gap of any case in this doc — are also the two largest values of mixing here: the states where the matrix punishes commitment hardest are the same ones where committing costs the most.

All fourteen cases, in full

The five above are the ones to read first. What follows is the complete set, Case 1 through Case 12 in order — the same five again, in their numbered place, plus the seven not covered above: a pure state for contrast, the fourth and last canonical support shape, a hedge so shallow that rounding it would misread the game, this project's own tackle rule producing the same duel by a different mechanism, the exact mirror of Case 2, a case worth discussing on its own (Case 9's twin structural finding, the three-lane mix), and a single-cell goal forced to mix anyway once movement noise is added — each with its board, its node-and-arrow graph, its exact 4×4 Q matrix, and its type label.

Case 1 (4, 0, 5, 0, 0)

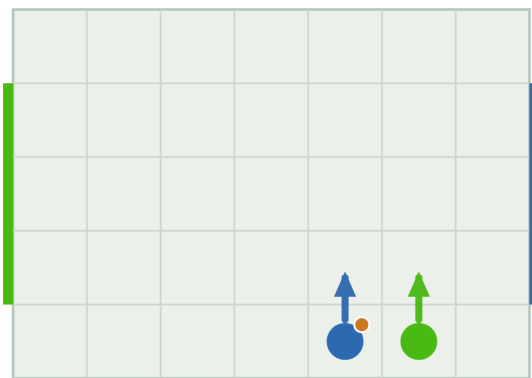
A pure state, for contrast. Saddle at U/U. The carrier plays U always; the defender answers U always; neither has any reason to deviate.

Type: pure equilibrium — not one of the 94 no-pure-saddle states; shown for contrast with everything else on this page.

Support: carrier {U} (100%) · defender {U} (100%)

Why indifferent: Carrier: $U=+0.234$ beats next-best $L=+0.161$ by 0.073; defender: $U=+0.234$ beats next-best $R=+0.271$ by 0.038. No tie on either side — that gap, not a coin flip, is what makes this state pure.

A pure state -- one safe cell, nothing to guess



pure equilibrium · $V = +0.234$

Q matrix -- (4, 0, 5, 0, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4×4 Q matrix as a node-and-arrow graph.

State	(4, 0, 5, 0, 0)
Carrier	player 0 at (4, 0)
Defender	player 1 at (5, 0)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.234446	0.316945	0.522973	0.271536
D	0.150878	0.211001	0.211001	0.222362
L	0.161034	0.182213	0.171623	0.189901
R	0.012406	-0.095109	0.057946	0.097586

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	pure	U 100.0%	U 100.0%
3 decimal(s)	pure	U 100.0%	U 100.0%
2 decimal(s)	pure	U 100.0%	U 100.0%
1 decimal(s)	pure	L 100.0%	U 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(4,1,5,1,0)	(4,1,5,0,0)	50%: (4,1,4,0,0) 50%: (4,1,5,0,0)	(4,1,6,0,0)
D	(4,0,5,1,0)	(4,0,5,0,0)	(4,0,5,0,0)	(4,0,6,0,0)
L	(3,0,5,1,0)	(3,0,5,0,0)	50%: (3,0,4,0,0) 50%: (3,0,5,0,0)	(3,0,6,0,0)
R	50%: (4,0,5,1,1) 50%: (5,0,5,1,0)	(4,0,5,0,1)	50%: (4,0,5,0,0) 50%: (4,0,5,0,1)	50%: (4,0,6,0,1) 50%: (5,0,6,0,0)

Case 2 (0, 1, 1, 1, 0)

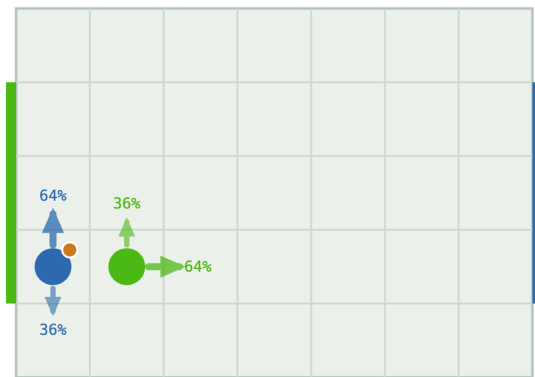
The primary example — the typical mix. Gap 0.0136. The carrier mixes U 63.5% / D 36.5%; the defender answers U 36.5% / R 63.5%. This is not one strange state among many — it is the **dominant pattern in the entire mixed-state set: 90 of the 94 no-pure-saddle states on the canonical board cross a vertical pair exactly like this one**, the carrier deciding which goal row to attack and the defender guessing which one. Everything that follows is either this same shape recurring (Case 7), or a genuinely different shape worth contrasting against it (Cases 3, 4, 5, 9, 10).

Type: forced mixed equilibrium — the reported 63.5%/36.5% split is the whole indifference class, not one point on a larger face.

Support: carrier {U, D} (63.5% / 36.5%) · defender {U, R} (36.5% / 63.5%)

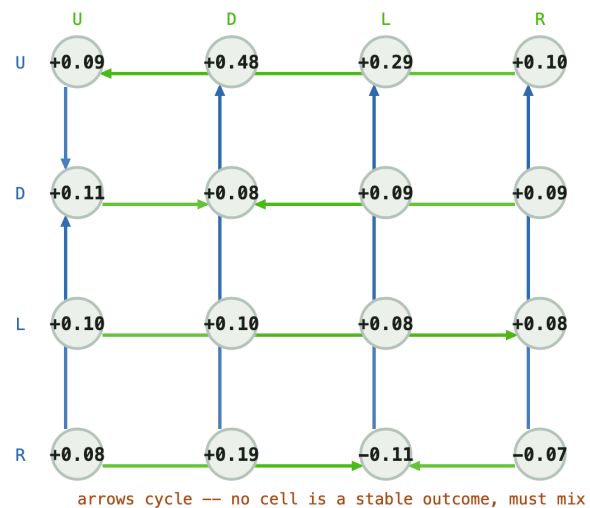
Why indifferent: Carrier $E[U] = E[D] = +0.094$, strictly above $E[L] = +0.091$ and $E[R] = -0.015$. Defender $E[U] = E[R] = +0.094$, strictly below $E[L] = +0.212$ and $E[D] = +0.332$.

The primary example: the typical mix



mixed equilibrium · $V = +0.094$

Q matrix -- (0, 1, 1, 1, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 1, 1, 1, 0)
Carrier	player 0 at (0, 1)
Defender	player 1 at (1, 1)

Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.085599	0.478297	0.285670	0.099198
D	0.109260	0.076330	0.085331	0.085599
L	0.103381	0.103381	0.084811	0.083738
R	0.082480	0.187381	-0.108448	-0.070696

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 63.5% / D 36.5%	U 36.5% / R 63.5%
3 decimal(s)	mixed	U 63.9% / D 36.1%	U 36.1% / R 63.9%
2 decimal(s)	mixed	U 66.7% / D 33.3%	U 33.3% / R 66.7%
1 decimal(s)	pure	L 100.0%	U 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(0,2,1,2,0)	(0,2,1,0,0)	50%: (0,2,0,1,0) 50%: (0,2,1,1,0)	(0,2,2,1,0)
D	(0,0,1,2,0)	(0,0,1,0,0)	50%: (0,0,0,1,0) 50%: (0,0,1,1,0)	(0,0,2,1,0)
L	(0,1,1,2,0)	(0,1,1,0,0)	(0,1,1,1,0)	(0,1,2,1,0)
R	50%: (0,1,1,2,1) 50%: (1,1,1,2,0)	50%: (0,1,1,0,1) 50%: (1,1,1,0,0)	50%: (0,1,1,1,0) 50%: (0,1,1,1,1)	50%: (0,1,2,1,1) 50%: (1,1,2,1,0)

Case 3 (1, 1, 1, 0, 1)

A clean L/R indifference example. Placed directly after the primary example on purpose: this is the cleanest demonstration of the intuition Case 2 only implies —

"when we move right and when we move left, there's an equal chance of me winning. That's why I am indifferent between the two."

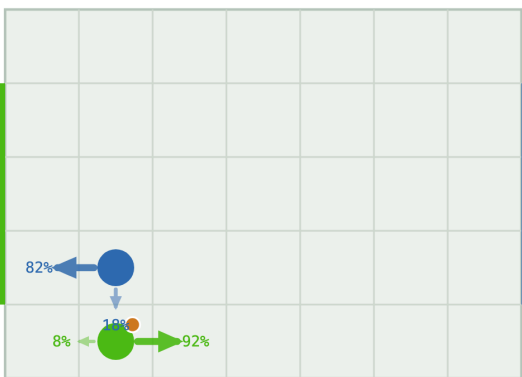
Gap 0.0445. The carrier's entire live option set is exactly $\{L, R\}$ — no vertical option survives at all, which is what makes this state (rather than Case 4, which crosses U/L) the cleanest match to that description: two actions genuinely equally good against the defender's own mix. **This is the rarer shape overall: only 4 of 94 no-pure-saddle states cross a purely horizontal pair instead of a vertical one.**

Type: forced mixed equilibrium — both sides' splits are the whole indifference class.

Support: carrier $\{L, R\}$ (7.7% / 92.3%) · defender $\{D, L\}$ (18% / 82%)

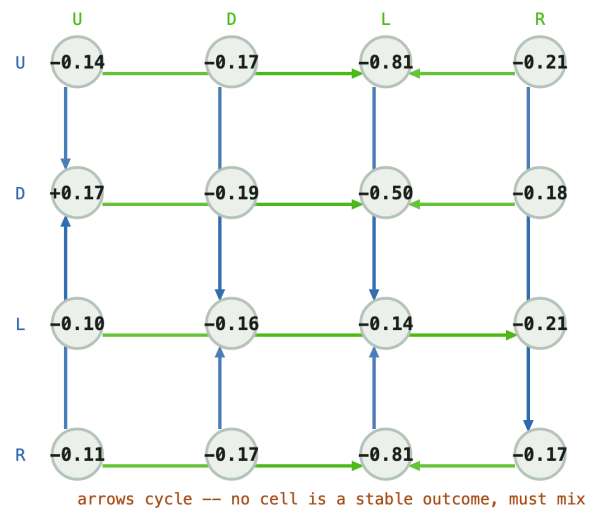
Why indifferent: Carrier $E[L]=E[R]=+0.206$, strictly above $E[D]=+0.162$ and $E[U]=+0.050$. Defender $E[D]=E[L]=+0.206$, strictly below $E[R]=+0.216$ and $E[U]=+0.257$.

A clean L/R indifference example



mixed equilibrium · $V = -0.206$

Q matrix -- (1, 1, 1, 0, 1)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(1, 1, 1, 0, 1)
Carrier	player 1 at (1, 0)
Defender	player 0 at (1, 1)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	-0.139279	-0.170790	-0.810000	-0.211001
D	0.173204	-0.185032	-0.499883	-0.180940
L	-0.099163	-0.156678	-0.141010	-0.211001
R	-0.109755	-0.170790	-0.810000	-0.166529

Where two of these cells actually come from — every entry is $Q(s, a_0, a_1) = \text{sum_outcomes } P(\text{outcome}) * (r_0 + \text{gamma} * V(\text{next}))$, real game.transitions() outcomes and the solver's own V, not the matrix solve itself. Two worked examples, $\text{gamma} = 0.9$:

$Q(U, U) = -0.139279$ — off both players' actual support, a bad joint action for player 0:

OUTCOME	P	NEXT	R0	V(NEXT)	P*(R0 + Γ *V(NEXT))
move first	0.5	(1, 2, 1, 1, 1)	0	-0.900000	-0.405000
move second	0.5	(1, 2, 1, 0, 0)	0	+0.590490	+0.265721
sum					-0.139279

$Q(D, U) = +0.173204$ — inside the defender's real 18% support, and notable because one of its two successors is the state itself:

OUTCOME	P	NEXT	R0	V(NEXT)	P*(R0 + Γ *V(NEXT))
move first	0.5	(1, 1, 1, 0, 0)	0	+0.590490	+0.265721
move second	0.5	(1, 1, 1, 0, 1) (= s itself)	0	-0.205592	-0.092516
sum					+0.173204

Neither cell has a per-step reward ($r_0 = 0$): the "win" objective only pays ± 1 on the step a goal is actually scored, so every non-terminal cell's value comes entirely from $\gamma * V(\text{next})$, discounted by whichever successor's own certified value it lands on — including, for $Q(D, U)$, this very state's own $V = -0.205592$, since D/U can return play to $(1, 1, 1, 0, 1)$ unchanged. (Cross-checked against `game.transitions()` and `result.values` directly in `tests/test_positions_verification.py`, not hand-derived.)

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0 (DEFENDER)	PLAYER 1 (CARRIER)
full (unrounded)	mixed, gap 0.0445	D 18.0% / L 82.0%	L 7.7% / R 92.3%
3 decimals	mixed	D 18.0% / L 82.0%	L 7.7% / R 92.3%
2 decimals	mixed	D 17.9% / L 82.1%	L 7.7% / R 92.3%
1 decimal	pure	L 100%	R 100%

Two decimals reproduces the exact mix to within a percentage point. One decimal manufactures a pure saddle — value exactly -0.2000 , defender pure L, carrier pure R — that is not actually there: it is an artifact of the rounding, not a feature of the game. There is no single mathematically privileged rounding precision; this table exists so that question is checkable rather than assumed one way or the other (`soccer_nash.numerics.rounding_diagnostic`).

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x_0, y_0, x_1, y_1, b) . Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	50%: (1,2,1,1,1) 50%: (1,2,1,0,0)	(1,2,1,0,1)	(1,2,0,0,1)	(1,2,2,0,1)
D	50%: (1,1,1,0,0) 50%: (1,1,1,0,1)	(1,1,1,0,1)	50%: (1,1,0,0,1) 50%: (1,0,0,0,1)	50%: (1,1,2,0,1) 50%: (1,0,2,0,1)
L	50%: (0,1,1,1,1) 50%: (0,1,1,0,0)	(0,1,1,0,1)	(0,1,0,0,1)	(0,1,2,0,1)
R	50%: (2,1,1,1,1) 50%: (2,1,1,0,0)	(2,1,1,0,1)	(2,1,0,0,1)	(2,1,2,0,1)

Why not defend by moving right instead? The natural reading of this state — player 1 has the ball right next to player 0 — is "close the distance," which on this board means R, not L. That reading misses one board detail: `goal_rows = (1, 2, 3)` on a 5-tall board, so row **y=0**, where the carrier currently stands, is not a scoring row at all — reaching $x=-1$ from $y=0$ does not score. The carrier is not a live threat yet; it first has to climb into a goal row before a leftward run means anything, and the cell it needs is (0, 1), one step from the defender's own current cell (1, 1) by playing L. The defender's job here is to deny that *future* cell, not to react to the carrier's *current* one.

The cost of getting this backwards is not subtle. Taking the worst case over the carrier's reply to each pure defender action, straight from the Q matrix above:

DEFENDER PLAYS	WORST CASE FOR DEFENDER	CARRIER'S BEST REPLY	SUCCESSOR STATE
U	-0.810	L	(1,2,0,0,1) -- defender at (1,2), corner wide open
R	-0.810	L	(2,1,0,0,1) -- defender moves away, same open corner
L	-0.211	R	(0,1,0,0,1) -- defender now sits on the corner itself

U and R are not merely worse than L, they are close to the worst possible outcome on this board — a goal is worth ± 1 , and -0.81 under $\gamma=0.9$ is what "the opponent is about to score" looks like in this game's own value scale. Playing L is not a retreat: the defender ends up standing on (0, 1), exactly the cell the carrier would otherwise have walked into uncontested.

Once that cell is covered, the carrier's own best reply shifts away from L toward R (92.3% of the mix) — not because R is a good attacking move on its own, but because L only pays off if $(0, 1)$ is left open, and the defender's 82% weight on L makes sure it usually is not.

This entire case depends on the goal convention. Everything above relies on row $y=0$ (where the carrier starts) not being a goal row under this project's `goal_rows = (1, 2, 3)`. Under a full-boundary goal (`goal_rows = (0, 1, 2, 3, 4)`, every row goal-eligible) this exact state is **pure, not mixed**: the carrier's L wins in two moves regardless of the defender's reply, all four of the defender's actions tie at the same value, and there is no guessing game left to play at all — see [generalize.md](#) for the numbers. `goal_rows = (1, 2, 3)` is Littman's own convention and this repo's default, confirmed as the better choice for this project, but Case 3 specifically — not just the aggregate mixed-state count — would need to be re-derived under a different one.

Case 4 (0, 0, 1, 1, 0)

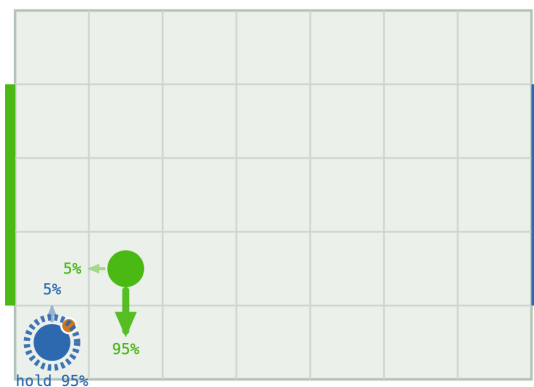
The corner duel. The board position sketched at the meeting: the carrier at (0, 0) — the back corner, right against its own goal — and the defender diagonally adjacent at (1, 1). This is the closest match, coordinate for coordinate, to that sketch. Read the carrier's split with the corner in mind: at (0, 0), L is a wall no-op — it clamps straight back onto the same cell, physically identical to D — so the 95.4% is not the carrier “escaping across the back line,” it is the carrier **holding its ground** in the corner. Its only genuine escape is straight up the sideline (U, 4.6%); the defender, one cell away, is guessing whether the carrier bolts or stays put. **Gap 0.0121; the mix is lopsided but genuine** — the corner leaves little room, but not zero.

Type: degenerate equilibrium face (carrier) — D ties the reported L split exactly (see “why indifferent”), which here is not an LP coincidence: D and L are the *same physical no-op* from (0, 0), so any split of that 95.4% between the two labels is an equally valid equilibrium; only the U/hold split (4.6%/95.4%) is actually forced.

Support: carrier {U, hold} (4.6% / 95.4%, printed as L) · defender {D, L} (94.9% / 5.1%, both real moves)

Why indifferent: Carrier $E[U] = +0.078$, above $E[R] = -0.066$; $E[D] = E[L] = +0.078$ too, guaranteed rather than coincidental since D and L are the same wall-clamped hold from this corner — any split of that 95.4% between the two labels is an equally valid equilibrium, not just the 0%/95.4% split shown. Defender $E[D] = E[L] = +0.078$, below $E[R] = +0.086$ and $E[U] = +0.109$.

The corner duel -- (0,0) and (1,1)



mixed equilibrium · $V = +0.078$

Q matrix -- (0, 0, 1, 1, 0)

	U	D	L	R
U	+0.10	+0.10	-0.41	+0.08
D	+0.11	+0.08	+0.10	+0.09
L	+0.11	+0.08	+0.10	+0.09
R	+0.11	-0.08	+0.11	+0.09

arrows cycle -- no cell is a stable outcome, must mix

Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 0, 1, 1, 0)
Carrier	player 0 at (0, 0)
Defender	player 1 at (1, 1)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.103381	0.103381	-0.407594	0.083738
D	0.109260	0.076330	0.100850	0.085599
L	0.109260	0.076330	0.100850	0.085599
R	0.112055	-0.075068	0.112055	0.088459

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 4.6% / L 95.4%	D 94.9% / L 5.1%
3 decimal(s)	mixed	U 4.7% / L 95.3%	D 95.0% / L 5.0%
2 decimal(s)	mixed	U 3.8% / L 96.2%	D 96.2% / L 3.8%
1 decimal(s)	pure	L 100.0%	D 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(0,1,1,2,0)	(0,1,1,0,0)	50%: (0,1,1,1,0) 50%: (0,0,0,1,1)	(0,1,2,1,0)
D	(0,0,1,2,0)	(0,0,1,0,0)	(0,0,0,1,0)	(0,0,2,1,0)
L	(0,0,1,2,0)	(0,0,1,0,0)	(0,0,0,1,0)	(0,0,2,1,0)
R	(1,0,1,2,0)	50%: (1,0,1,1,0) 50%: (0,0,1,0,1)	(1,0,0,1,0)	(1,0,2,1,0)

Case 5

(0, 0, 2, 0, 0)

A **genuine 3-action mix**. Six cells from goal, as far as this board allows, with the defender close enough to threaten three different rows at once: no single row is safely better than the other two, so three actions are simultaneously undominated on both sides. **This is one of only 4 states on the whole canonical board with support shape (3,3)** — mixing is not always a 50/50 split between two actions. (Entropy 1.109 bits, the richest mix on this page, if a single summary number is wanted — but the three-way split itself is the point.)

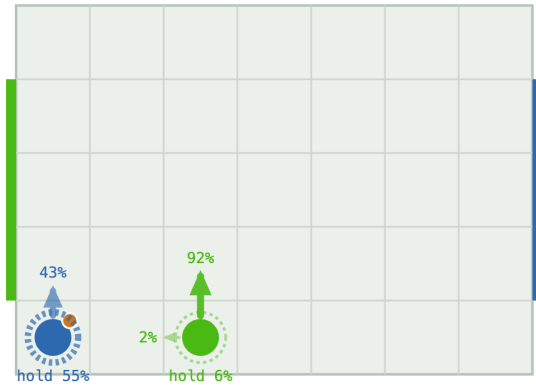
Both players are pinned against an edge here, so read the raw action letters with that in mind: the carrier sits at (0, 0), so L is a wall no-op — it clamps straight back to (0, 0), physically identical to D — which is why the board figure draws that 54.7% as a dashed **hold** ring, not a leftward arrow. The three genuinely distinct carrier choices are *attack up* (U, 43.3%), *hold ground* (D/L combined, 54.7%), and *sneak right* (R, 1.9%) — still a real three-way mix, just not the three-arrows picture the raw letters would suggest. The defender at (2, 0) has the same issue in miniature: D is its own wall no-op (5.9%, folded into its own hold ring), alongside real moves U (91.9%) and L (2.3%).

Type: degenerate equilibrium face (carrier) — a fourth, unweighted carrier action ties the reported three-way split exactly (see “why indifferent”), so the printed split is one point on a larger face, not a forced ratio. Here that tie is not a coincidence of the LP: D and L are the *same physical action* from (0, 0), so $E[D]=E[L]$ is a mechanical certainty, not a numerical accident. The defender's own split is a genuine forced mix.

Support: carrier {U, L, R} (43.3% / 54.7% / 1.9%, L a wall hold) · defender {U, D, L} (91.9% / 5.9% / 2.3%, D a wall hold)

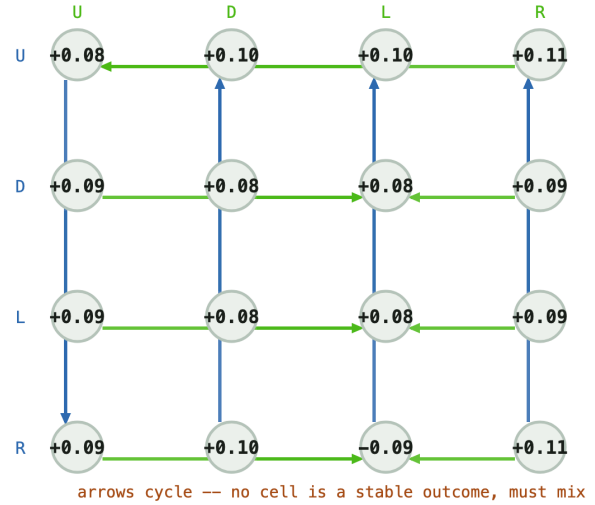
Why indifferent: Carrier $E[U]=E[L]=E[R]=+0.0848$ (the three actions weighted) — and $E[D]=+0.0848$ too: **all four actions tie exactly**, and $E[D]=E[L]$ here is guaranteed before any LP runs, since D and L are the same wall-clamped no-op from (0, 0). The printed 43.3%/0%/54.7%/1.9% split is one point on a whole face of equally-good carrier mixes, not a uniquely forced ratio. Defender $E[U]=E[D]=E[L]=+0.0848$, below $E[R]=+0.097$.

A genuine 3-action mix, not a 2-cycle



mixed equilibrium · $V = +0.085$

Q matrix -- (0, 0, 2, 0, 0)



arrows cycle -- no cell is a stable outcome, must mix

Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 0, 2, 0, 0)
Carrier	player 0 at (0, 0)
Defender	player 1 at (2, 0)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.083738	0.095109	0.103381	0.110220
D	0.085599	0.076363	0.076330	0.085599
L	0.085599	0.076363	0.076330	0.085599
R	0.088459	0.095109	-0.088213	0.110220

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 43.3% / L 54.7% / R 1.9%	U 91.9% / D 5.9% / L 2.3%
3 decimal(s)	mixed	U 46.0% / L 52.0% / R 2.0%	U 91.2% / D 6.9% / L 1.9%
2 decimal(s)	mixed	U 33.3% / L 66.7%	U 66.7% / D 29.8% / L 3.5%
1 decimal(s)	pure	L 100.0%	L 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x_0, y_0, x_1, y_1, b) . Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	$(0, 1, 2, 1, 0)$	$(0, 1, 2, 0, 0)$	$(0, 1, 1, 0, 0)$	$(0, 1, 3, 0, 0)$
D	$(0, 0, 2, 1, 0)$	$(0, 0, 2, 0, 0)$	$(0, 0, 1, 0, 0)$	$(0, 0, 3, 0, 0)$
L	$(0, 0, 2, 1, 0)$	$(0, 0, 2, 0, 0)$	$(0, 0, 1, 0, 0)$	$(0, 0, 3, 0, 0)$
R	$(1, 0, 2, 1, 0)$	$(1, 0, 2, 0, 0)$	50%: $(1, 0, 2, 0, 0)$ 50%: $(0, 0, 1, 0, 1)$	$(1, 0, 3, 0, 0)$

Case 6

(1, 1, 2, 0, 1)

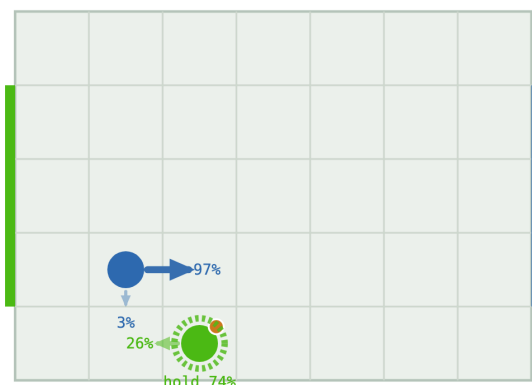
A near-pure hedge, where rounding would lie. Gap 0.0043. The defender plays R 97.5% / D 2.5% — entropy only 0.169 bits, a hedge so lopsided that reading 0.975 as 1.0 would be a real rounding mistake: the gap is certified and real, twelve times smaller than a 0.1 rounding grid would resolve. The carrier's own mix is more balanced (D 73.9% / L 26.1%) — it is the defender's near-pure hedge that makes this case worth including, not the carrier's. Read the carrier's D with the board in mind: the carrier sits at (2, 0), the bottom row, so D is a wall no-op — 73.9% of the time it holds position rather than moving down, and the other 26.1% it slides left (L). Unlike Case 4 or 5, this tie is a genuine strategic indifference between two *different* physical options (hold vs. move), not two labels for the same action.

Type: forced mixed equilibrium — lopsided, but neither side has a tied outsider action.

Support: carrier {hold, L} (73.9% / 26.1%, printed as D/L) · defender {D, R} (2.5% / 97.5%, both real moves)

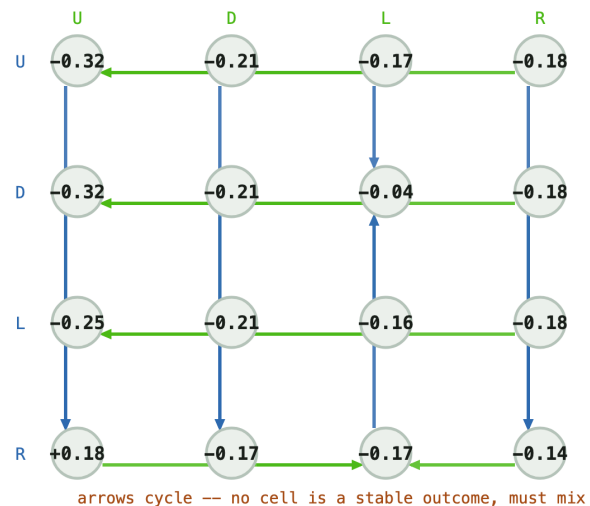
Why indifferent: Carrier $E[D]=E[L]=+0.168$, above $E[R]=+0.136$ and $E[U]=-0.166$ — D here is the carrier holding its position at the bottom edge, genuinely tied with a real move left. Defender $E[D]=E[R]=+0.168$, below $E[L]=+0.197$ and $E[U]=+0.201$.

A near-pure hedge -- where rounding would lie



mixed equilibrium · $V = -0.168$

Q matrix -- (1, 1, 2, 0, 1)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(1, 1, 2, 0, 1)
Carrier	player 1 at (2, 0)
Defender	player 0 at (1, 1)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	-0.316945	-0.211001	-0.170790	-0.182213
D	-0.316945	-0.211001	-0.044962	-0.182213
L	-0.247069	-0.211001	-0.156678	-0.182213
R	0.178022	-0.166529	-0.170790	-0.135158

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	D 2.5% / R 97.5%	D 73.9% / L 26.1%
3 decimal(s)	mixed	D 2.4% / R 97.6%	D 74.1% / L 25.9%
2 decimal(s)	pure	R 100.0%	D 76.5% / L 23.5%
1 decimal(s)	pure	R 100.0%	D 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(1,2,2,1,1)	(1,2,2,0,1)	(1,2,1,0,1)	(1,2,3,0,1)
D	(1,0,2,1,1)	(1,0,2,0,1)	50%: (1,0,2,0,0) 50%: (1,1,1,0,1)	(1,0,3,0,1)
L	(0,1,2,1,1)	(0,1,2,0,1)	(0,1,1,0,1)	(0,1,3,0,1)
R	50%: (2,1,2,0,0) 50%: (1,1,2,1,1)	(2,1,2,0,1)	(2,1,1,0,1)	(2,1,3,0,1)

Case 7 (5, 1, 6, 1, 1)

The typical mix, mirrored. The board's left-right / role-swap symmetry maps Case 2 exactly onto this state. The values negate to machine precision:

$V(\text{case 2}) = +0.094235$
 $V(\text{mirror}) = -0.094235$
 $\text{sum} = +2.78\text{e-}17$ (zero, to floating-point noise)

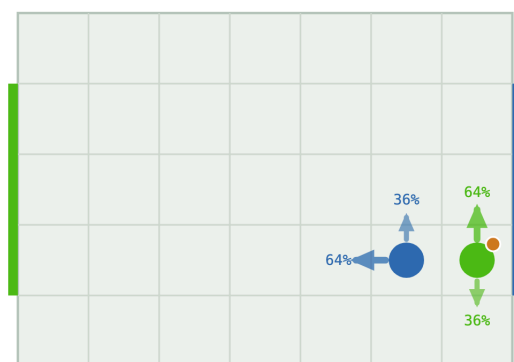
Direct evidence for the repetition claim above: this is not a new shape, it is Case 2's identical mix reappearing, exactly mirrored, wherever the same relative configuration recurs.

Type: forced mixed equilibrium — Case 2's exact classification, mirrored.

Support: carrier $\{U, D\}$ (63.5% / 36.5%) · defender $\{U, L\}$ (36.5% / 63.5%)

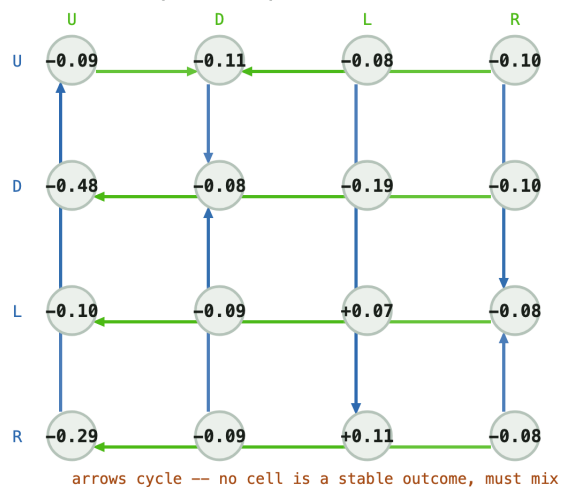
Why indifferent: Carrier $E[U]=E[D]=+0.094$, above $E[R]=+0.091$ and $E[L]=-0.015$. Defender $E[U]=E[L]=+0.094$, below $E[R]=+0.212$ and $E[D]=+0.332$ — Case 2's exact chain, mirrored.

The typical mix, mirrored -- same shape, value negated



mixed equilibrium · $V = -0.094$

Q matrix -- (5, 1, 6, 1, 1)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(5, 1, 6, 1, 1)
Carrier	player 1 at (6, 1)
Defender	player 0 at (5, 1)

Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	-0.085599	-0.109260	-0.082480	-0.103381
D	-0.478297	-0.076330	-0.187381	-0.103381
L	-0.099198	-0.085599	0.070696	-0.083738
R	-0.285670	-0.085331	0.108448	-0.084811

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 36.5% / L 63.5%	U 63.5% / D 36.5%
3 decimal(s)	mixed	U 36.1% / L 63.9%	U 63.9% / D 36.1%
2 decimal(s)	mixed	U 33.3% / L 66.7%	U 66.7% / D 33.3%
1 decimal(s)	pure	U 100.0%	R 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(5,2,6,2,1)	(5,2,6,0,1)	50%: (5,2,5,1,1) 50%: (5,2,6,1,0)	(5,2,6,1,1)
D	(5,0,6,2,1)	(5,0,6,0,1)	50%: (5,0,5,1,1) 50%: (5,0,6,1,0)	(5,0,6,1,1)
L	(4,1,6,2,1)	(4,1,6,0,1)	50%: (4,1,5,1,1) 50%: (4,1,6,1,0)	(4,1,6,1,1)
R	50%: (5,1,6,2,1) 50%: (6,1,6,2,1)	50%: (5,1,6,0,1) 50%: (6,1,6,0,1)	50%: (5,1,6,1,0) 50%: (5,1,6,1,1)	(5,1,6,1,1)

Case 8 (2, 3, 3, 3, 1), 5×4 board

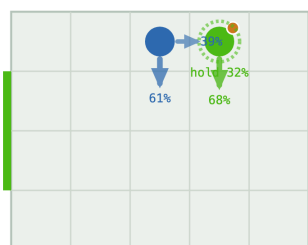
This project's own tackle rule. Every other case here gets its coin flip from Littman's random move order. This one is solved under the tackle rule (defender commits to a challenge that wins the ball with fixed probability or bounces off) instead — a mechanism with nothing to do with move order at all. It produces the same kind of duel (gap 0.0223): the carrier mixes U 32.1% / D 67.9%, the defender mixes D 60.7% / R 39.3%. Different cause, same structure. The carrier sits at (3, 3), the top row of this 5×4 board, so U is a wall no-op: 32.1% of the time it holds its ground rather than pushing up, and 67.9% of the time it drives down (D), a real move.

Type: forced mixed equilibrium — same classification as the move-order-driven cases above, under a completely different mechanism.

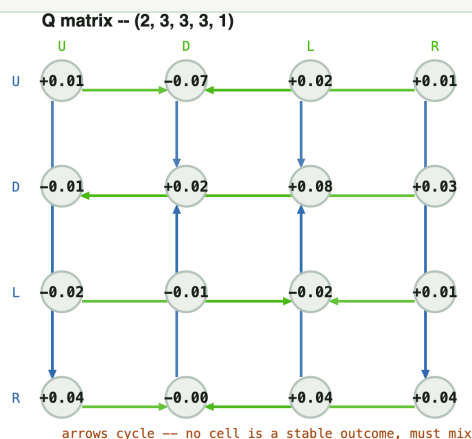
Support: carrier {hold, D} (32.1% / 67.9%, printed as U/D) · defender {D, R} (60.7% / 39.3%, both real moves)

Why indifferent: Carrier $E[U] = E[D] = -0.013$, above $E[R] = -0.031$ and $E[L] = -0.065$ — U here is the carrier holding at the top edge, genuinely tied with a real move down. Defender $E[D] = E[R] = -0.013$, below $E[L] = +0.011$ and $E[U] = +0.045$.

This project's own tackle rule -- a different mechanism, same duel



mixed equilibrium · $V = +0.013$



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4×4 Q matrix as a node-and-arrow graph.

State	(2, 3, 3, 3, 1)
Carrier	player 1 at (3, 3)
Defender	player 0 at (2, 3)

Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.011944	-0.072350	0.018582	0.007062
D	-0.005242	0.022044	0.080389	0.025575
L	-0.015879	-0.008641	-0.018582	0.006356
R	0.041869	-0.000279	0.041869	0.039427

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	D 60.7% / R 39.3%	U 32.1% / D 67.9%
3 decimal(s)	mixed	D 60.9% / R 39.1%	U 31.9% / D 68.1%
2 decimal(s)	mixed	D 57.1% / R 42.9%	U 28.6% / D 71.4%
1 decimal(s)	pure	D 100.0%	D 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(2,3,3,3,1)	(2,3,3,2,1)	(2,3,3,3,0)	(2,3,4,3,1)
D	(2,2,3,3,1)	(2,2,3,2,1)	(2,2,2,3,1)	(2,2,4,3,1)
L	(1,3,3,3,1)	(1,3,3,2,1)	(1,3,2,3,1)	(1,3,4,3,1)
R	50%: (3,3,4,3,0) 50%: (2,3,3,3,1)	50%: (3,3,4,3,0) 50%: (2,3,3,2,1)	50%: (3,3,4,3,0) 50%: (2,3,3,3,1)	50%: (3,3,4,3,0) 50%: (2,3,4,3,1)

Case 9 (0, 2, 1, 2, 0)

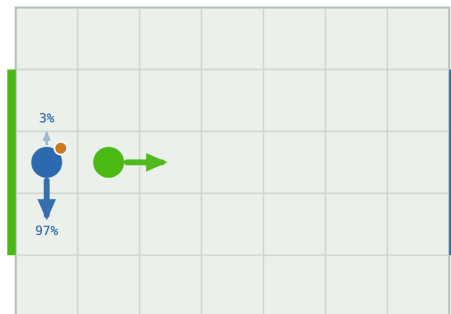
The asymmetric mix — the sharpest LP-degeneracy example. Support (2, 1)
 — gap 0.0095. The odd one out: the carrier's equilibrium still mixes two actions (U 2.6% / D 97.4%), but the **defender's** equilibrium is a single fixed move (R, 100%). Every other case on this page is a symmetric duel — both players genuinely guessing. Here only one side is.

Type: pure-tied / fractional LP output (defender) — the defender's official policy is printed pure R (100%), but D ties it exactly: the choice of R over D is exactly as arbitrary as a printed fractional split. The sharpest example on this page that a fractional LP output is not automatically a strategically required mixed strategy.

Support: carrier {U, D} (2.6% / 97.4%) · defender {R} (100%)

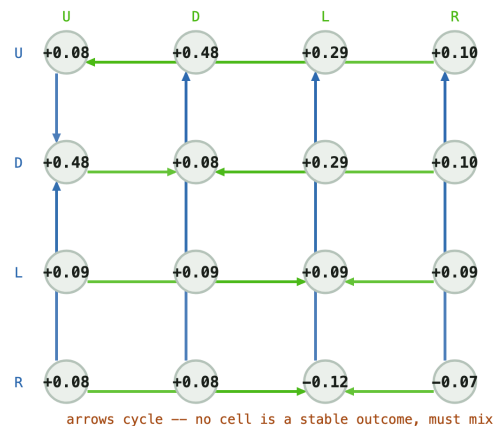
Why indifferent: Carrier $E[U] = E[D] = +0.095$, above $E[L] = +0.093$ and $E[R] = -0.072$ (worked out fully below). The defender's official policy is pure R (100%), but $E[R] = +0.095$ and $E[D] = +0.095$ tie **too**: the defender's own choice of R over D is exactly as arbitrary as the carrier's U/D split, it just isn't printed as a percentage because the solver put all the weight on one side.

The asymmetric mix -- a tie, not a forced mix, dressed the same



mixed equilibrium · $V = +0.095$

Q matrix -- (0, 2, 1, 2, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 2, 1, 2, 0)
Carrier	player 0 at (0, 2)
Defender	player 1 at (1, 2)
Rows	Player 0
Columns	Player 1

Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball
---------------	--

	U	D	L	R
U	0.084811	0.478297	0.290839	0.095109
D	0.478297	0.084811	0.290839	0.095109
L	0.093043	0.093043	0.085599	0.093043
R	0.082303	0.082303	-0.122276	-0.071591

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 2.6% / D 97.4%	R 100.0%
3 decimal(s)	mixed	U 2.5% / D 97.5%	R 100.0%
2 decimal(s)	mixed	U 5.0% / D 95.0%	R 100.0%
1 decimal(s)	pure	D 100.0%	R 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(0,3,1,3,0)	(0,3,1,1,0)	50%: (0,3,0,2,0) 50%: (0,3,1,2,0)	(0,3,2,2,0)
D	(0,1,1,3,0)	(0,1,1,1,0)	50%: (0,1,0,2,0) 50%: (0,1,1,2,0)	(0,1,2,2,0)
L	(0,2,1,3,0)	(0,2,1,1,0)	(0,2,1,2,0)	(0,2,2,2,0)
R	50%: (0,2,1,3,1) 50%: (1,2,1,3,0)	50%: (0,2,1,1,1) 50%: (1,2,1,1,0)	50%: (0,2,1,2,0) 50%: (0,2,1,2,1)	50%: (0,2,2,2,1) 50%: (1,2,2,2,0)

Why the carrier still needs two actions against a defender who never varies: against the defender's pure R, column R reads U 0.095, D 0.095, L 0.093, R -0.072 — **U and D are exactly tied**, both strictly better than L or R. Nothing in the payoffs favours one over the other, so any split of U/D is an equally valid best reply; the solver's LP reports one particular split (2.6% / 97.4%), not a probability forced by the game the way Case 3's 7.7% / 92.3% is. This is the sharpest example on this page of a general fact worth stating

plainly: a fractional LP output is not automatically a strategically required **mixed strategy**. A tie between two rows can look identical, in the printed policy, to a genuine forced mix, but it is a different phenomenon — no best-reply cycle is involved, and either pure U or pure D alone would also be a valid equilibrium reply to the defender's fixed R.

Case 10

(0, 2, 2, 2, 1)

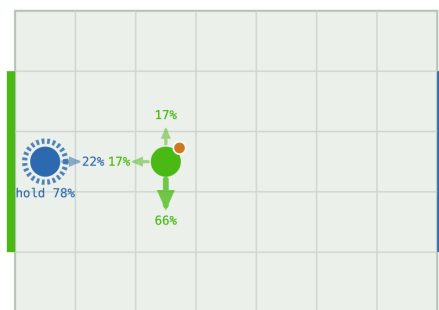
The three-lane mix. Support (3, 2) — gap 0.0699, the deepest gap of any state on this page (fifteen times deeper than the shallowest hedge in Case 6). This is the fourth and last canonical support shape: the carrier genuinely needs three live actions, while the defender only ever needs two. Combined with Case 9, this settles a structural question the certificates make checkable rather than assumed: **the defender never needs strictly more actions than the carrier**, across all 94 no-pure-saddle states on the canonical board. The defender sits at (0, 2), the left edge, so its reported majority action L (77.9%) is a wall no-op — it holds position rather than moving left; the genuine move in its support is R (22.1%).

Type: degenerate equilibrium face (defender) — D ties the reported {L, R} split exactly (see “why indifferent”), so the defender is really indifferent among three actions even though the LP only split weight over two — and one of those two, L, is itself the wall-clamped hold, not a real move. The carrier's own split is a genuine forced mix.

Support: carrier {U, D, L} (16.9% / 66% / 17.1%) · defender {hold, R} (77.9% / 22.1%, printed as L/R)

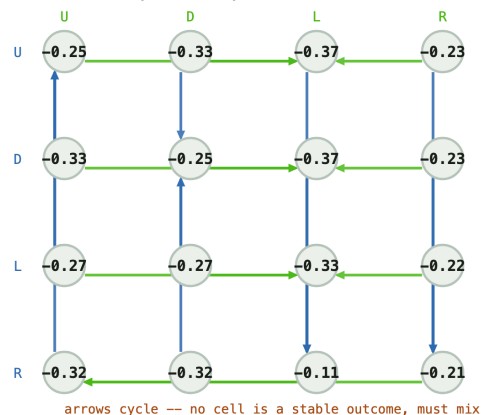
Why indifferent: Carrier $E[U]=E[D]=E[L]=+0.282$, above $E[R]=+0.217$ — a genuine three-way tie. The defender's official support is {L, R} at +0.282, below $E[U]=+0.322$ — but $E[D]=+0.282$ too: the defender is really indifferent among *holding* (L), moving down (D), and moving right (R), even though the LP only split weight over two of them.

The three-lane mix -- support (3,2), the deepest gap on this page



mixed equilibrium · $V = -0.282$

Q matrix -- (0, 2, 2, 2, 1)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 2, 2, 2, 1)
Carrier	player 1 at (2, 2)
Defender	player 0 at (0, 2)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	-0.247069	-0.330151	-0.366481	-0.228062
D	-0.330151	-0.247069	-0.366481	-0.228062
L	-0.271536	-0.271536	-0.330151	-0.219944
R	-0.316945	-0.316945	-0.109966	-0.205131

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	L 77.9% / R 22.1%	U 16.9% / D 66.0% / L 17.1%
3 decimal(s)	mixed	L 78.1% / R 21.9%	U 17.6% / D 65.4% / L 17.0%
2 decimal(s)	mixed	L 77.8% / R 22.2%	U 11.1% / D 70.4% / L 18.5%
1 decimal(s)	mixed	U 28.6% / D 28.6% / R 42.9%	U 42.9% / D 42.9% / L 14.3%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(0,3,2,3,1)	(0,3,2,1,1)	(0,3,1,2,1)	(0,3,3,2,1)
D	(0,1,2,3,1)	(0,1,2,1,1)	(0,1,1,2,1)	(0,1,3,2,1)
L	(0,2,2,3,1)	(0,2,2,1,1)	(0,2,1,2,1)	(0,2,3,2,1)
R	(1,2,2,3,1)	(1,2,2,1,1)	50%: (1,2,2,2,0) 50%: (0,2,1,2,1)	(1,2,3,2,1)

Case 11 (4, 4, 5, 4, 0), deterministic

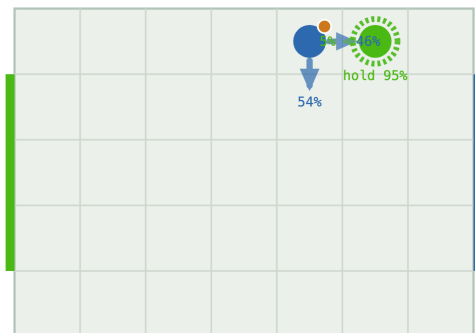
Mixing forced by reward alone, not transition. An advanced case, illustrating a fundamentally different mechanism from the four above. Every case so far gets its coin flip from a stochastic transition — the random move order. This one has **no transition stochasticity at all**: the mix comes entirely from scoring="territory", a dense per-step reward for the ball in the opponent's final third, layered on top of the ordinary win/loss score. Gap 0.0653 — close to a fair coin (entropy 0.996 bits), forced purely by coupling the reward to both players' actions. **Mixing does not require stochastic transitions; it can come from the payoff structure alone.** The defender sits at (5, 4), the top edge, so its reported majority action U (95.1%) is a wall no-op: it holds position 95.1% of the time and slides left (L) the other 4.9%.

Type: forced mixed equilibrium — reward-coupling forces a genuine indifference on both sides, the same as a transition-coupled matching-pennies core.

Support: carrier {D, R} (53.8% / 46.2%, both real moves) · defender {hold, L} (95.1% / 4.9%, printed as U/L)

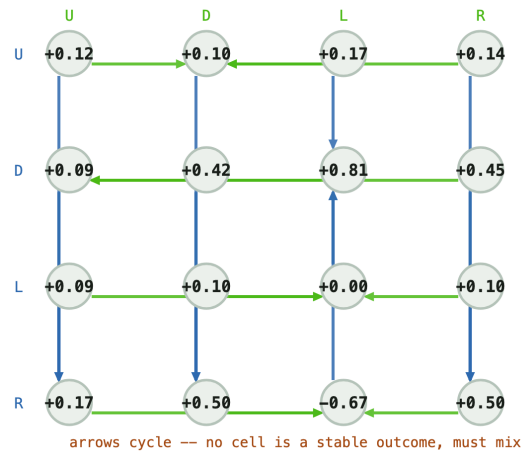
Why indifferent: Carrier $E[D] = E[R] = +0.129$, above $E[U] = +0.119$ and $E[L] = +0.090$. Defender $E[U] = E[L] = +0.129$, below $E[D] = +0.455$ and $E[R] = +0.473$ — U here is the defender holding at the top edge, tied with a real move left.

Mixing forced by reward alone -- zero transition randomness



mixed equilibrium · $V = +0.129$

Q matrix -- (4, 4, 5, 4, 0)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(4, 4, 5, 4, 0)
Carrier	player 0 at (4, 4)
Defender	player 1 at (5, 4)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.116139	0.104525	0.169846	0.135972
D	0.094073	0.415483	0.814500	0.450000
L	0.094073	0.104525	0.000000	0.104525
R	0.169846	0.500000	-0.670721	0.500000

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	D 53.8% / R 46.2%	U 95.1% / L 4.9%
3 decimal(s)	mixed	D 53.8% / R 46.2%	U 95.1% / L 4.9%
2 decimal(s)	mixed	D 53.8% / R 46.2%	U 94.9% / L 5.1%
1 decimal(s)	mixed	D 56.2% / R 43.8%	U 93.8% / L 6.2%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(4,4,5,4,0)	(4,4,5,3,0)	(4,4,5,4,1)	(4,4,6,4,0)
D	(4,3,5,4,0)	(4,3,5,3,0)	(4,3,4,4,0)	(4,3,6,4,0)
L	(3,4,5,4,0)	(3,4,5,3,0)	(3,4,4,4,0)	(3,4,6,4,0)
R	(4,4,5,4,1)	(5,4,5,3,0)	(5,4,4,4,1)	(5,4,6,4,0)

Case 12 (1, 3, 1, 4, 1)

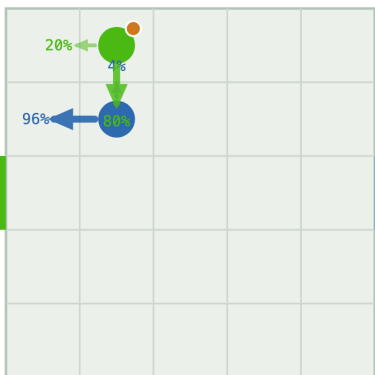
Movement slip on a single-cell goal. The sharpest possible contrast with Case 1. A single-cell goal under deterministic move order is **pure across every configuration tested in this project** — that is the whole goal-width result; a board-size-free proof remains open. Add $\text{slip}=0.15$ (each player independently takes a uniform-random move instead of its chosen one, 15% of the time, regardless of position) to that exact board, and **52 mixed states appear where there were 0**. Gap 0.0172 — a genuine, certified mix, not noise. The goal-width switch is a property of Littman's move-order coin specifically; a coin that lands on every square regardless of the goal forces mixing everywhere, independent of goal width.

Type: forced mixed equilibrium — movement noise produces the same genuine indifference as the move-order coin, on a board shape that is otherwise always pure.

Support: carrier {D, L} (80.2% / 19.8%) · defender {U, L} (3.9% / 96.1%)

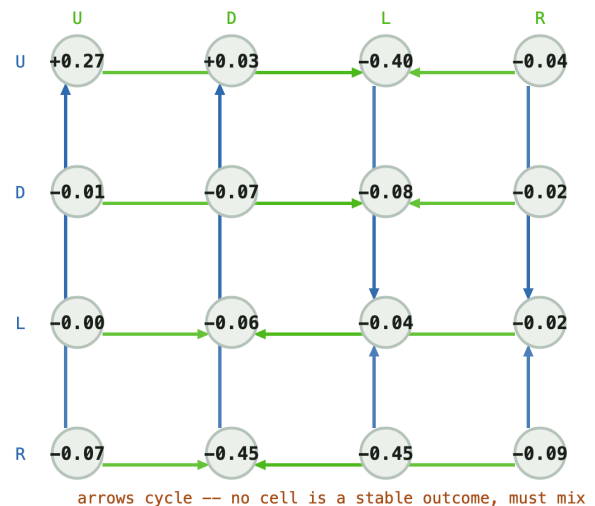
Why indifferent: Carrier $E[D]=E[L]=+0.053$, above $E[R]=+0.016$ and $E[U]=-0.010$. Defender $E[U]=E[L]=+0.053$, below $E[D]=+0.072$ and $E[R]=+0.450$.

Movement slip -- a single-cell goal, pure until now



mixed equilibrium · $V = -0.053$

Q matrix -- (1, 3, 1, 4, 1)



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(1, 3, 1, 4, 1)
Carrier	player 1 at (1, 4)

Defender	player 0 at (1, 3)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.273301	0.031430	-0.396398	-0.039469
D	-0.005960	-0.068847	-0.083433	-0.020201
L	-0.001041	-0.056711	-0.039473	-0.015300
R	-0.069053	-0.449923	-0.449665	-0.091802

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 3.9% / L 96.1%	D 80.2% / L 19.8%
3 decimal(s)	mixed	U 4.0% / L 96.0%	D 80.2% / L 19.8%
2 decimal(s)	mixed	U 4.4% / L 95.6%	D 80.0% / L 20.0%
1 decimal(s)	mixed	U 20.0% / L 80.0%	D 80.0% / L 20.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

Shown: the intended successor before movement slip is applied (slip=0 isolates it, for readability). The actual game has slip=0.15; running python scripts/positions.py on this state prints the full noisy distribution — up to 16 outcomes per cell, each intended move landing on its target around 79% of the time and spreading over nearby cells the rest. See [generalize.md](#).

P0 \ P1	U	D	L	R
U	(1,3,1,4,0)	(1,4,1,3,0)	(1,4,0,4,1)	(1,4,2,4,1)
D	(1,2,1,4,1)	(1,2,1,3,1)	(1,2,0,4,1)	(1,2,2,4,1)
L	(0,3,1,4,1)	(0,3,1,3,1)	(0,3,0,4,1)	(0,3,2,4,1)
R	(2,3,1,4,1)	(2,3,1,3,1)	(2,3,0,4,1)	(2,3,2,4,1)

Case 13 (1, 1, 3, 1, 1)

Template 3, the last un-cased matching-pennies shape.

docs/templates.md reduces the 94 no-pure-saddle states to 8 canonical geometric templates; Cases 2, 3, 4, 5, 9, and 10 above turn out to already be instances of six of them (recomputed from each state's own geometry key, not assumed). This case and Case 14 are the two that were missing: this one is **template 3** (16 of the 94 states) — the defender two cells ahead of the carrier, on the same goal row. Gap 0.0419.

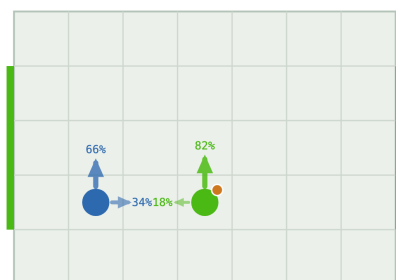
Player 1 carries. The carrier's live moves are U (step to the next goal row) and L (drive straight down the row at the defender); the defender answers with U (cover the row above) or R (hold the lane it is already in). Same crossing shape as Case 2 and Case 7, one row further apart.

Type: forced mixed equilibrium — both sides' splits are the whole indifference class. The nearest unweighted action is the defender's L, 0.0064 above its support's tied value — a real gap, not a rounding-level tie the way Case 9's D is.

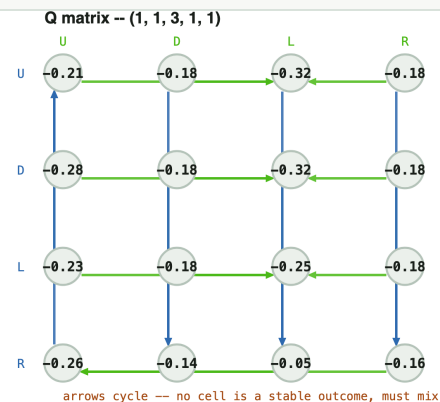
Support: carrier {U, L} (82.3% / 17.7%) · defender {U, R} (65.6% / 34.4%)

Why indifferent: Carrier $E[U] = E[L] = +0.225$, above $E[D] = +0.166$ and $E[R] = +0.175$. Defender $E[U] = E[R] = +0.225$, below $E[D] = +0.289$ and $E[L] = +0.231$.

Template 3 -- defender two ahead on the goal row, the last un-cased shape



mixed equilibrium · $V = -0.225$



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(1, 1, 3, 1, 1)
Carrier	player 1 at (3, 1)

Defender	player 0 at (1, 1)
Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	-0.205131	-0.182213	-0.316945	-0.182213
D	-0.283411	-0.182213	-0.316945	-0.182213
L	-0.228062	-0.182213	-0.247069	-0.182213
R	-0.262831	-0.135158	-0.049186	-0.162618

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 65.6% / R 34.4%	U 82.3% / L 17.7%
3 decimal(s)	mixed	U 65.6% / R 34.4%	U 82.2% / L 17.8%
2 decimal(s)	mixed	U 65.6% / R 34.4%	U 84.4% / L 15.6%
1 decimal(s)	pure	L 100.0%	R 100.0%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(1,2,3,2,1)	(1,2,3,0,1)	(1,2,2,1,1)	(1,2,4,1,1)
D	(1,0,3,2,1)	(1,0,3,0,1)	(1,0,2,1,1)	(1,0,4,1,1)
L	(0,1,3,2,1)	(0,1,3,0,1)	(0,1,2,1,1)	(0,1,4,1,1)
R	(2,1,3,2,1)	(2,1,3,0,1)	50%: (2,1,3,1,0) 50%: (1,1,2,1,1)	(2,1,4,1,1)

Case 14 (0, 2, 2, 2, 0)

Template 8, the rarest shape on the board. Template 8 covers only 2 of the 94 no-pure-saddle states — the rarest of the eight, a wall-clamped variant of template 4's 2x1 shape (carrier two actions, defender one): the defender two cells ahead of the carrier, same goal row, with the carrier pinned against its own back wall (x=0). Gap 0.0178.

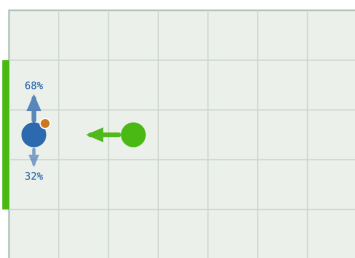
Player 0 carries, already at the leftmost column. Its only live moves are U/D (there is nowhere further left to go — L is a wall no-op); the defender's reported reply is pure L (hold its lane).

Type: pure-tied / fractional LP output (defender) — the defender's printed policy is pure L (100%), but U ties it exactly (both +0.1034): the defender's choice of L over U is as arbitrary as a printed fractional split, the same phenomenon as Case 9, just on the other canonical 2x1 template. The carrier's U/D split, in contrast, is a genuine forced tie — both strictly above L and R by a real margin.

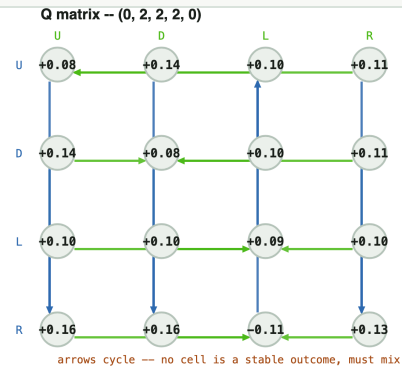
Support: carrier {U, D} (67.5% / 32.5%) · defender {L} (100%, tied with U)

Why indifferent: Carrier $E[U]=E[D]=+0.103$, above $E[L]=+0.086$ and $E[R]=-0.110$. The defender's official policy is pure L (+0.103), but $E[U]=+0.103$ too: D=+0.125 and R=+0.113 are both clearly worse for the defender, but the L/U choice is a coin flip the LP happened to call L.

Template 8 -- the rarest shape (2 of 94 states), a wall-clamped defender-pure tie



mixed equilibrium · $V = +0.103$



Left: where the players are, with each live action drawn as a probability arrow. Right: the full 4x4 Q matrix as a node-and-arrow graph.

State	(0, 2, 2, 2, 0)
Carrier	player 0 at (0, 2)
Defender	player 1 at (2, 2)

Rows	Player 0
Columns	Player 1
Payoff	Player 0's Q-value – fixed for every state, never reoriented by who has the ball

	U	D	L	R
U	0.083738	0.144198	0.103381	0.112892
D	0.144198	0.083738	0.103381	0.112892
L	0.099198	0.099198	0.085599	0.099198
R	0.155749	0.155749	-0.109966	0.132828

Rounding diagnostic — the same matrix, solved again after rounding it to each precision (not just checking whether classification flips):

PRECISION	CLASSIFICATION	PLAYER 0	PLAYER 1
full (unrounded)	mixed	U 67.5% / D 32.5%	L 100.0%
3 decimal(s)	mixed	U 68.3% / D 31.7%	L 100.0%
2 decimal(s)	mixed	U 66.7% / D 33.3%	L 100.0%
1 decimal(s)	pure	L 100.0%	D 66.7% / L 33.3%

Successor states, by joint action — rows = player 0, columns = player 1, fixed, same as the Q matrix above, each cell the resulting (x0, y0, x1, y1, b). Two lines in a cell means the outcome depends on who moves first (each 50%).

P0 \ P1	U	D	L	R
U	(0,3,2,3,0)	(0,3,2,1,0)	(0,3,1,2,0)	(0,3,3,2,0)
D	(0,1,2,3,0)	(0,1,2,1,0)	(0,1,1,2,0)	(0,1,3,2,0)
L	(0,2,2,3,0)	(0,2,2,1,0)	(0,2,1,2,0)	(0,2,3,2,0)
R	(1,2,2,3,0)	(1,2,2,1,0)	50%: (1,2,2,2,0) 50%: (0,2,1,2,1)	(1,2,3,2,0)

The mathematics behind all fourteen

A mixed equilibrium is exactly the strategy pair where every action in a player's support earns the **same expected payoff** against the opponent's own

mix — that equality is what "indifferent" means in Case 3, and it is also why the best-response graph cycles with no resting point: if one action in the support paid strictly more, that player would raise its weight on it, which is precisely the condition an equilibrium rules out. A best-reply cycle and mutual indifference are the same fact seen from two sides, not two different explanations. Case 9 shows the boundary of that story: indifference alone (a tie against a fixed opponent) can produce the same *symptom* — a fractional policy — without a cycle anywhere in the matrix.

Deterministic vs. stochastic transitions

Every one of the fourteen cases above is tagged by which kind of transition produced it, because that tag is the entire finding, not incidental detail:

CASE	BOARD / RULE	TRANSITION
1	canonical	deterministic outcome (the joint action happens to have a pure saddle)
2, 3, 4, 6, 7, 9, 10, 13	canonical, move_order="random"	stochastic — a coin decides who moves first
5	canonical, move_order="random"	stochastic, but the mix is forced by geometry (three lanes at once), not by the coin alone
8	this project's own tackle rule	stochastic — a duel resolves with probability tackle_prob
11	move_order="deterministic", scoring="territory"	deterministic transition, mixed anyway — the reward, not the transition, forces it
12	move_order="deterministic", slip=0.15	deterministic policy outcome, but movement itself is noisy
14	canonical, move_order="random"	stochastic — a wall-clamped variant of Case 9's asymmetric shape

Case 1 and Case 11 are the pair worth holding side by side: both have a **deterministic transition function** — no coin anywhere — and yet Case 1 is pure while Case 11 is mixed. The switch is not "is there randomness in the rules"; it is the collision/reward structure of that specific case (see docs/report.md §5's coinflip-vs-random-order comparison, which makes the same point the other way: adding *deterministic* randomness — coinflip, a fair coin whose outcome doesn't depend on the joint action — leaves every stage game pure). Randomness alone is neither necessary (Case 11 above) nor sufficient (coinflip) for mixing; the switch is move_order="random" specifically, because it is the only rule where the *same* joint action resolves differently depending on the coin, coupling the transition to the players' choices instead of merely adding noise to them.

On the live site, the Board Explorer makes this a direct comparison rather than a table: its board dropdown includes both **Canonical board — random move order** and **Canonical board — deterministic (A10-style)**, same 7x5 geometry, so switching between them holds everything else fixed. Its **Simulate** panel makes the consequence concrete — self-play the exact equilibrium from kickoff and the deterministic board settles into essentially all draws, while the random board produces a decisive, lopsided result, the same contrast a research group member's own site demoed live.

Does mixing depend on *where* on the board, or the *configuration*?

The configuration, not the raw location. docs/generalize.md tests this directly by sweeping board size, aspect ratio, and goal shape: the same switch — a goal wide enough that the defender cannot cover every scoring cell at once — keeps flipping mixing on, on boards from the smallest tested up to roughly 7,800 states, regardless of where that goal sits or how large the board is around it. docs/templates.md makes the "configuration" half precise: every one of the 94 no-pure-saddle states on the canonical board reduces, after the board mirror and a carrier-frame reorientation, to just **8 canonical templates**, and every one of them occurs at Manhattan distance 1 or 2 *between the two players* — never distance 3 or more, regardless of how far either player is from the goal itself.

So the honest answer has two parts: it is not "this square of the board is special" (raw location doesn't predict it — the same relative geometry recurs all over the board and across board sizes); it is "this relationship between the two players is special" (defender-to-carrier distance, plus a goal too wide to cover from one cell) — a property of the configuration that generalizes, not a property of any one place.

Minimax vs. always-left vs. random vs. best response

The same four policies the live Board Explorer's Simulate panel lets you play against each other, computed exactly here instead of by self-play sampling: **minimax** (the exact Nash policy), **always-left** (the simplest fixed opponent — play L every step, Littman's own baseline), **random** (uniform over the four

actions), and **best response** — not a fixed policy at all, but whatever exactly maximises against the *other* side's actual policy that game. Every cell is **player 0's exact expected discounted goal difference from kickoff** (scoring="rate", gamma=0.9, the same canonical 7x5 board every case above uses), row = player 0's policy, column = player 1's:

PLAYER 0 \ PLAYER 1	MINIMAX	ALWAYS-LEFT	RANDOM	BEST RESPONSE
minimax	+0.1323	+0.6246	+0.5350	+0.1323
always-left	+0.0000	+0.0000	+0.0000	-0.0000
random	-0.1423	-0.2160	+0.0059	-0.4445
best response	+0.1323	+0.7637	+0.6842	+0.1323

Three things this table makes exact, not just plausible:

1. **Minimax vs. best-response equals minimax vs. minimax.** Both read +0.1323 — the defining property of a Nash policy: the best anyone can do against it, *knowing its exact mixing probabilities*, is no better than what minimax already guarantees itself. That is what "unexploitable" means, made numeric instead of asserted.
2. **Always-left is not merely bad, it is invariant.** Every one of its cells reads 0.0000 (to rounding) regardless of the opponent, because a player that only ever plays L at its own kickoff cell is a wall-clamped no-op — it never moves, never releases the ball, and never scores, so there is nothing for either side's policy to react to. It is the degenerate case of "a fixed action has a fixed counter": here the counter is simply "do nothing back."
3. **Every fixed (non-best-responding) policy has a ceiling a best-responder reaches exactly.** Minimax's ceiling against best-response is its own value (+0.1323, unchanged from minimax-vs-minimax); random's collapses to its worst cell (-0.4445) once the opponent plays the exact best reply instead of a random guess. This is docs/tournament.md's "every deterministic offense has a perfect defense" (Littman's Table 3, MR/MM vs. QR/QQ collapsing to 0.0% won against a trained challenger), reproduced here as the general **minimax-vs-everything** matrix rather than one row of it.

Reproduces with python `scripts/positions_policy_matrix.py`,
which writes `experiments/positions_policy_matrix.csv`.

Reproduce with `python scripts/positions.py (make positions)` — writes `figures/gallery/positions.svg` and prints every exact matrix (via `NashQIteration.run_exact()`'s direct linear solve, not the iterative approximation to it), policy, support, and successor-state coordinate table shown here. `python scripts/pure_vs_mixed_exploit.py` reproduces the pure-vs-mixed exploitability table above. `python scripts/degeneracy.py` reproduces the forced/degenerate/pure-tied classification behind every "Type" label above. `python scripts/positions_policy_matrix.py` reproduces the minimax/always-left/random/best-response matrix above. Every number on this page, including the last two sections, is re-verified against the solver directly in `tests/test_positions_verification.py` (`pytest -m slow`). Full project: github.com/Charith-Reddy-Pareddy/soccer-markov-nash.